Senior DataOps Engineer – TRC Anne-Maria – Scentre Group v2 Interviewed on 15/7/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DataOps Engineer /Senior DevOps Engineer /Senior Data Analyst /Python/R Developer/Clouds/Linux Administrator /AI/ML/BI/SharePoint/Dynamics 365 CRM Consultant /EDW - Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DataOps Engineer/Senior DevOps Engineer /Python/R Developer/AI/ML /SP/D365 CRM Consultant/EDW
Responsibilities:

- Build and operate secure, scalable and optimal data pipelines for Commission's GCP platform using Python, SQL by following 'DataOps' processes and principles. Heavy focus on systems automation and CI/CD to deploy everything.
- Designed and implemented a **Google Cloud Platform EDW** using a serverless Lambda architecture. Data is ingested via Pub/Sub layer and processed in real-time streaming layer using Cloud Functions. Cloud Dataflow performs periodic batch processing to update the data warehouse stored in BigQuery under serving layer, as the single source of truth for analytics.
- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code (IaC) tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency. Or leveraged MuleSoft Anypoint Platform's capabilities for automated provisioning & configuration within the cloud environment
- Designed a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure**: Azure DevOps
    - **AWS**: AWS CodePipeline/CodeBuild/CodeDeploy
    - **GCP**: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Implemented a scalable data warehouse leveraging GCP's BigQuery service, facilitating efficient querying and analysis of extensive datasets through dimensional data modeling.
- Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.
- **Big data wrangling**: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- Utilized distributed file systems like HDFS (Hadoop Distributed File System) or GCS (Google Cloud Storage) to efficiently store and access terabytes of data across multiple machines.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.
- Developed a secure, local AI system for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Built robust & secure GCP Landing Zones with Terraform, ensuring compliance with HIPAA, GDPR, and PCI DSS standards, and cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
- Developed and maintained CI/CD pipelines for a variety of applications using GCP DevOps, DevSecOps towards AIOps.

- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
- Utilized Cloud Composer or Airflow for orchestrating workflows and advanced scheduling within GCP Dataflow. These tools integrate well with Dataflow and offer a native GCP experience.
- Employ Dataform for data pipeline definition as code, managing SQL code with version control, data asset definition, and pipeline documentation. Utilized Dataform to manage the "what" of the pipelines, while Cloud Composer/Airflow handles the "when" and "how" of execution. Used agile methodologies to deliver projects on time and within budget.
- Deploying GKE as a managed Kubernetes service for running containerized applications on GCP, managing Kubernetes clusters with automatic scaling, load balancing, and health checks to ensure high availability and scalability of applications.
- Developed and maintained IaC scripts using Python and Terraform for provisioning and managing cloud resources, enhancing scalability and flexibility of applications. Developing Data APIs (REST, GraphQL, Graph API, gRPC)

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior DataOps Engineer/Senior DevOps Engineer /Python/R Developer /AI/ML /BI/SP Consultant /EDW
Responsibilities:

Built and secured hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Utilize Cloud Composer or Airflow for orchestrating workflows and advanced scheduling within GCP Dataflow.

Configure dbt for BigQuery by specifying the project ID, dataset, and other connection details in the profiles.yml file

Automated data pipelines using Airflow and Dataform, ensuring efficient data processing and delivery.

Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.

Collaborated with business analysts and product owners to translate data requirements into actionable solutions, driving data-driven decision-making. Utilized Talend for ETL/ELT processes, achieving streamlined data integration.

Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation templates or Terraform. Utilized AWS Control Tower for centralized management of multiple AWS accounts, enabling parallel processing for various projects.

Develop and implement comprehensive unit tests to validate the functionality and performance of the application.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

Leveraged Cloud Build to automate building and testing of data pipeline code. Utilized Cloud Storage to securely store data pipeline code and artifacts. Cloud Monitoring tracked the performance of data pipeline

Implement automated monitoring with Prometheus and Grafana to collect metrics for all RBA's data pipelines.

Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.

Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.

Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to

Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items

Seamlessly execute automated tests within GCP utilizing Selenium, Cypress, or Appium. Leverage GCP's scalable infrastructure like Compute Engine or Kubernetes Engine for parallel test execution, ensuring effortless adaptation to growing business needs.

Conduct non-functional acceptance testing, develop test scripts, and execute test cases for system enhancements.

Leveraged SAP Data Services and SAP Business Objects to deliver robust ETL, Reporting, and Analytics solutions, showcasing in-depth understanding of these processes.

Migrated on-premise applications to the Google Cloud Platform (GCP) cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos, and CloudEndure Migration to GKE environment.

Employing Dataform for data pipeline orchestration, SQL code management with version control, data asset definition, testing and deployment environment creation, and pipeline documentation.

Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab CI/CD, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.
Deploy and manage microservices-based applications using Docker CRE and Kubernetes on GCP GKE clusters.
Administered and monitored both Windows and Linux-based systems, ensuring smooth operation of critical applications.
Integrate security, code analysis, and quality checks into the SDLC automation using SonarQube, Snyk, and OWASP ZAP.
Monitor and optimize system and application performance using Stackdriver, Datadog, Splunk and New Relic.
Engineered AI-driven ML systems using Python and machine learning frameworks (PyTorch, scikit-learn)
Used MLOps for the entire machine learning application lifecycle, encompassing development, deployment, & maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production environments.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DataOps Engineer/Snr DevOps Engineer/Python/R Developer /AI/ML/SharePoint /K2 Developer/DBA
Responsibilities:

Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
Leverage Kubespray with its community-maintained extensions and Ansible to automate Kubernetes cluster deployment on AWS and Azure clouds. Then, utilize Kustomize for managing Kubernetes applications deployed on Amazon EKS and Azure AKS, enabling consistent configurations and simplified deployments across both cloud providers.
Migrated on-premises data platforms to GCP, leveraging services like BigQuery, Dataflow, and Cloud Storage.
Engineered and operated data platforms on Google Cloud Platform, utilizing services such as BigQuery, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.
Designing and developing complex data pipelines in a data analytics environment, dealing with large datasets
Utilize Cloud Composer or Airflow for orchestrating workflows and advanced scheduling within GCP Dataflow.
Configure dbt for BigQuery by specifying the project ID, dataset, and other connection details in the profiles.yml file
Automated data pipelines using Airflow and Dataform, ensuring efficient data processing and delivery.
Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to
As a product development team member, utilized tools like Selenium, Cypress, and Appium to run automated tests seamlessly within GCP. Leveraging GCP's scalable infrastructure like Compute Engine or Kubernetes Engine ensures parallel test execution and effortless adaptation to growing business needs.
Proficient in designing and implementing data solutions on AWS, utilizing services like EC2 for data compute, EBS for storage, RDS for databases, and S3 for data lakes, alongside expertise in Spark/PySpark SQL for big data querying and analysis, and Impala for low-latency analytical workloads.
Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.
Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.
Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
Leveraging Dataform for managing intricate data pipelines with dependency management, data quality checks, and data lineage tracking. Using Cloud Build to automate diverse CI/CD tasks such as building, testing, and deploying applications to various platforms like Kubernetes, Cloud Functions, and App Engine.
Leveraging BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment.
Conducted non-functional acceptance testing, developed test scripts, and executed non-functional test cases for Data and Integration systems including the deployment of application and data pipeline releases while ensuring operational process and procedural documentation are maintained professionally.
Completed design, data models, loading approaches and source to target mapping documents.
Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
Created a mapping document that defines how the source data fields correspond to the target data fields.
Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
Implemented data transformation using Cloud Workflows and Dataproc Notebooks, writing BigQuery SQL scripts or utilizing PySpark for transformations, including data cleaning, enrichment, and aggregation.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,

Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases

- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Industry: DataOps Engineer /DevOps Engineer/Clouds/Linux Administrator/SharePoint Consultant
Responsibilities:

- Implemented a **3-tier lambda architecture EDW on Azure, combining batch and real-time processing**. An **ingestion layer** handles data intake, followed by a **batch processing layer** using HDInsight and PySpark for scalable data transformation. The **real-time analysis layer** leverages Azure Stream Analytics on Data Streams for near real-time insights. Additionally, Databricks SQL offers interactive querying on Data Lake Storage data.
- Migrated on-premises data platforms to GCP, leveraging services like BigQuery, Dataflow, and Cloud Storage.
- Deploying GKE as a managed Kubernetes service for running containerized applications on GCP, managing Kubernetes clusters with automatic scaling, load balancing, and health checks to ensure high availability and scalability of applications.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all UC's data pipelines.
- Leveraging BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment.
- Created and implemented ETL processes to extract, transform, and load data into data warehouses and data lakes.
- Implement the infrastructure as code using Terraform and Google Cloud Deployment Manager.
- Implemented secure CI/CD pipelines using Terraform, Vault, Artifactory, and GitLab for various applications.
- Implemented scripting solutions (Python, Bash) to automate routine tasks and improve operational efficiency.
- Provided technical support to internal users and troubleshoot data access and quality problems.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Demonstrated proficiency in scripting languages such as Python/PowerShell, Ansible, Puppet and Go for automation.
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Seamlessly execute automated tests within GCP utilizing Selenium, Cypress, or Appium/Appgee. Leverage GCP's scalable infrastructure like Compute Engine or Kubernetes Engine for parallel test execution, ensuring effortless adaptation to growing business needs. Monitored and maintained system performance, identified and resolved performance bottlenecks.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DataOps Engineer /DevOps Engineer /ML /SharePoint Consultant
Responsibilities:

- Performed data analysis using SQL and reporting tools to support business intelligence initiatives.
- Developed and maintained data reports and dashboards for various stakeholders.
- Cleaned and transformed raw data to ensure data quality and consistency.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Shell, Ansible and Puppet
- Proficient in scripting and programming languages, including Python and Ruby.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Configured and managed DNS, DHCP, Active Directory, Exchange Server, SharePoint Server, SQL Server, and
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DataOps Engineer /Data Analyst /BI Developer /Senior Systems Engineer /DWH /DBA

Responsibilities:

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Data Engineer /Data Analyst /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code,

Zhi Ming ZHANG - Australian Citizen File Ref: ZMZITILJP003
1303 / 29 Hunter St.| Parramatta 2150 | NSW Mobile: 0402-997-679 | mingjp@gmail.com

Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior DevOps Engineer – AKE-DPHI-4/7/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Senior Integration Developer/Software Engineer/Snr Data Engineer/Java/.Net/Python Developer /Clouds/Linux Administrator /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 Devloper /EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr DevOps Engineer/Snr Integration Developer/Software Engineer/Java/.Net/Python/AI/ML/SP/D365 Developer
Responsibilities:

- Develop high-performance, innovative code and software/system solutions for our customers.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365/Power BI/Power Automate & Power App)
- **Integration Solution Developer with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365.
- **Technical Proficiency:** Developed and managed cloud-based applications using Azure Functions for serverless execution and Azure Kubernetes Service for container orchestration in microservice architecture.
- Developed Azure Functions using C#, Node.js, Python, Java, or PowerShell for serverless functionalities.
- **Specific Skill with API Management**: Designed and implemented serverless microservices using Azure Functions triggered by HTTP requests. Utilized Azure API Management to expose these functions as secure and manageable APIs.
- **Action-oriented with Achievements**: Implemented over 20 Azure Functions (C#, Node.js, etc.) for various backend tasks, improving API response times by 25%.
- Used Azure API Management to secure & expose these functions as public APIs, resulting in a 30% increase in API usage.
- Automated data exchange between disparate systems using MuleSoft Anypoint Studio 7 and Mule 4 Platform. Implemented MuleSoft DataWeave (or MEL) for data transformation, resulting in a 25% improvement in data processing speed.
- Designed a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Leveraged Microsoft Power Apps to design user interfaces and workflows for data management. Utilized Power Automate for automating data-related tasks, such as data collection, manipulation, and integration with various data sources.
- **Innovative thinker**: Automates platform delivery with agile DevOps and new technologies
- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code (IaC) tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency. Or leveraged MuleSoft Anypoint Platform's capabilities for automated provisioning & configuration within the cloud environment

- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
  - **Azure**: Azure DevOps
  - **AWS**: AWS CodePipeline/CodeBuild/CodeDeploy
  - **GCP**: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- **Problem-Solving Abilities**: Advanced skills in analysing complex issues and developing practical, innovative solutions.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Commitment to Continuous Learning**: Committed staying ahead with latest technologies advancements & best practices.
- Leveraged Azure Cosmos DB, a schema-agnostic NoSQL database, to efficiently store and manage unstructured vector data like social media posts, videos, images, and audio.
- Implemented Azure Cognitive Search for native vector search methods, enabling fast and accurate search functionality. (Optional) Additionally, integrated libraries like Preflexity and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro for advanced search capabilities.
- **Customer Focus**: Strong orientation towards understanding and meeting customer needs and requirements.
- Developed medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- **Proficient in Documentation**: Proactive in generating reports, business cases, and written advice with minimal supervision.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior DevOps/Data Engineer/Snr Integration Developer/Java/.Net/Python/AI/ML/SP/Power BI Developer
Responsibilities:

Develop high-performance, innovative code and software/system solutions for our customers.
Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Technical Proficiency: Designed and deployed scalable and reliable software applications using Azure Logic Apps.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Skilled Full-Stack Engineer leveraging React/TypeScript for frontend, Java/APIs for backend development and expertise in AWS cloud. Proficient in integration tools like MuleSoft and J2EE for seamless data flow using RESTful web services.
Developed Azure Functions using C#, Node.js, Python, Java, or PowerShell for serverless functionalities.
Designed and implemented serverless microservices using Azure Functions triggered by HTTP requests. Utilized Azure API Management to expose these functions as secure and manageable APIs.
Implemented over 20 Azure Functions (C#, Node.js, etc.) for various backend tasks, improving API response times by 25%.
Used Azure API Management to secure & expose these functions as public APIs, resulting in a 30% increase in API usage.
Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.
Integrated DevSecOps practices with Infrastructure as Code (IaC) security principles to automate security configurations and ensure consistent security posture across deployments.
Stakeholder Collaboration: Excellent communication skills to engage effectively with internal teams, customers &vendors
Autonomous Work Ethic: Capability to work independently, represent the department, and drive forward-thinking solutions.
Project Management Skills: Experience in managing complex systems integrations and maintaining service integrity while implementing new initiatives.
Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline. The pipeline will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
Leveraged cutting-edge machine learning techniques to automate the creation and refinement of infrastructure diagrams based on natural language descriptions. This empowered DevOps engineers to gain deeper insights and optimize complex systems more efficiently.
Work cross-functionally with internal teams, customers, and vendors to develop solutions in line with our strategic priorities.

Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
Used BizTalk Server for legacy system integration by defining data structures (schemas), mapping data formats, utilizing adapters for communication protocols (e.g., FILE, REST), & orchestrating message flows (optional) for complex scenarios.
Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.
Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.
Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.
Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.
Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Docker,, GitLab/GitHub Actions, & Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
Proficient in Documentation: Proactive in generating reports, business cases, and written advice with minimal supervision.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr DevOps Engineer/Sr Integration Developer/Sr Data Engineer /.Net/Python ML/SPoint/K2 Developer
Responsibilities:
Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
Technical Proficiency: Designed and deployed scalable and reliable software applications on Microsoft Azure using Azure App Service and Virtual Machines. Implemented robust access controls and data protection for Azure environments leveraging Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center.
Medallion Architecture with Data Lake in Azure:
Batch Layer: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
Serving Layer: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
Real-time Layer: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
Leveraged Azure Blueprints to design and implement secure Virtual Private Clouds (VPCs) and integrate cloud workloads, focusing on Windows-based systems in M365 environments (O365/MS Teams/SharePoint 2016/Dynamics 365 /Power BI /Power Automate, etc.). This IPaaS solution ensures seamless data exchange and application communication.
Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
Unleash automation magic! Combine code (Python, Go) for custom logic with Workato's pre-built integrations to seamlessly automate complex workflows across your business, simplifying even the most intricate processes.
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Docker, GitLab/GitHub Actions, and Jenkins
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
Optimized code for ML projects using data structures, algorithms, and software architecture knowledge.
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.

Applied natural language processing techniques such as text preprocessing, sentiment analysis and topic modeling using NLTK (Natural Language Toolkit) and Gensim.
Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: DevOps Engineer/Integration Developer /Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- **Technical Proficiency:** Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,

Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Compile timely reports, business cases and other forms of written advice for leadership; maintain accurate records.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Systems Integration Developer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Responsible for integration development process, ensuring adherence to standards and documentation of requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior DevOps Engineer (Python Web Application Productionisation) – Macquarie Group – Immerge – 13/6/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter –Interviewed on 26/6/24
Senior DevOps Engineer (Python Web Application Productionisation)/Java/Python Developer/Snr Data Engineer / Integration Solution Architect /Salesforce/Mulesoft/AI/ML/DL/BI/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr DevOps Engineer (Python Web Application Productionisation)/Python/AI/ML/DL/SP/D365 CRM Developer
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code (IaC) tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Collaborate with development and operations teams to streamline the deployment process of our Python Web applications.
- Full-stack engineer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask/Django) to ensure seamless data flow using RESTful web services. Develop Python web application components using frameworks like Flask or Django.
- Skilled in building microservices applications using Python & JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with ACQSC's enterprise services layer, databases, and host systems.
- Onboard the application to security tools to ensure compliance with industry standards and best practices.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation templates or Terraform. Utilized AWS Control Tower for centralized management of multiple AWS accounts, enabling parallel processing for various projects.
- Develop and implement comprehensive unit tests to validate the functionality and performance of the application.
- Integrate the application with Ping SSO for seamless single sign-on authentication.
- **Configure AWS SSO (Service Provider)**: In the AWS account, add the Ping Identity application through SAML metadata (EC2, S3, Lambda, RDS) in the AWS SSO console. Map user attributes from Ping Identity to AWS (IAM).
- **Application Configuration (Client-side)**: For web apps hosted on AWS (EC2, Lambda), use the AWS SDK to integrate with AWS SSO and retrieve user information. For mobile/desktop apps, utilize Ping Identity's SDK for the SSO flow.
- Verify successful integration with Ping SSO by accessing the application through the AWS SSO portal. Users should be verified and automatically logged in to services like EC2, S3, Lambda, or RDS.
- Create technical documentation to facilitate knowledge sharing and maintain system integrity.
- Design and implement secure and scalable CI/CD pipelines leveraging AWS services:
- **Automated Builds**: Utilize AWS CodeBuild for automated build, test, and deployment processes.

- **Scalable Infrastructure**: Integrate with RDS for databases, S3 for storage, & Bitbucket repositories for code management.
- **Optional Manual Provisioning**: EC2 can be used for additional manual server deployments if needed.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- Troubleshoot and resolve issues related to deployment, configuration, and performance.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer/Mulesoft/Java/Python/R/AI/ML/DL/BI/SharePoint Developer /Snr Data Engineer
Responsibilities:

Hybrid Cloud Architect: Established secure connections between AWS, GCP, and Azure cloud environments for data transfer. Utilized infrastructure-as-code tools (Terraform/Ansible, or Puppet) for automated and consistent provisioning.

Built secure and scalable GCP Landing Zones using Infrastructure as Code (IaC) with Terraform or Cloud Build. Utilized GCP Cloud IAM for centralized governance and management of multiple GCP projects, enabling efficient parallel processing for various projects.

Full-stack engineer skilled in Angular/TypeScript for frontend development, Python/Flask or Django for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask/Django) to ensure seamless data flow using RESTful web services.

Develop Python web application components using frameworks like Flask or Django.

Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with RBA's enterprise services layer, databases, and host systems.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.

Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.

Implemented automated security scans (SAST, DAST) into the GitLab CI pipeline, enforced granular access controls using roles, and managed secrets securely. Regular audits ensured compliance with industry standards like PCI-DSS or SOC 2.

- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: `template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']`

Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.

Automation and CI/CD: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
Streamlined Terraform configuration management through the strategic use of the merge function to consolidate settings from diverse sources.
Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
CI/CD for Integration Solutions: Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).
Data Transformation Expertise: Utilized MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.
Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.
Optimized CI/CD pipelines with DevOps tools: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
Production-Ready Machine Learning Models: Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
CI/CD and DevSecOps Automation: Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
AI-Powered Pattern Recognition: Used Artificial Neural Networks to model complex patterns and forecast potential issues.
High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.
Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Integration Solution Architect/Snr DevOps Engineer /Java/Python/Salesforce/Mulesoft/SP/K2 Developer
Responsibilities:
Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.
Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
Full-stack engineer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask/Django) to ensure seamless data flow using RESTful web services. Develop Python web application components using frameworks like Flask or Django.
Designed and maintained robust CI/CD pipelines using platform-specific tools:

7. **Azure: Azure DevOps**
8. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
9. **GCP: Cloud Build/Cloud Deploy**

Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
Building microservices applications using Python & JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.
Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
Developed and deployed secure, RESTful Python APIs for web applications using frameworks like Flask or FastAPI. These APIs can integrate with MLOps pipelines for data access, model predictions, or other machine learning tasks.
Monitor and optimise the performance, scalability, and reliability of the application in a cloud environment.
Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.
Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics CRM 2016.
Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.
Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
Design and implement secure & scalable CI/CD pipelines using AWS CodeBuild, RDS & S3 with Bitbucket repositories.
Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
Developed and deployed a MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.
Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.
CI/CD and DevSecOps Automation (AWS): Built and maintained CI/CD pipelines for various applications, employing DevSecOps practices using industry-standard tools (CodeBuild/GitHub) & following best practices for secure development

- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Automation:** Created and maintained automated scripts using Python or Perl programming, along with Bash or Shell scripting, to streamline development and operational processes.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Collaborate with development and operations teams to streamline the deployment process of our Python Web applications.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- Troubleshoot and resolve issues related to deployment, configuration, and performance.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.

- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.

- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

IT Developer – Hicksons Lawyer – Engage – 10/6/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Snr IT Developer (Java/.Net/Dynamics 365/Power Platform/Salesforce/Mulesoft /Nintex K2 workflows) /Clouds Administrator /Integration Solution Architect/Snr DevOps/Data Engineer/Python/AI/ML/SPoint Developer –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr IT Developer (Java/.Net /Dynamics 365/Power Platform)/Snr DevOps Engineer/Python/AI/ML/SP Developer
Responsibilities:

- Designed and implemented secure Azure landing zones for cloud workloads using Azure Blueprints, with a focus on M365 environments (O365, MS Teams, SharePoint 2019, Dynamics 365, Power BI, Power Automate, Power Apps etc.).
- Research, design, develop, test, and maintain innovative and engaging digital solutions that streamline the firm's operations and enhance client services. This includes developing Salesforce/MuleSoft workflows for tasks such as Electronic health records workflow, automating lead nurturing processes, integrating customer data across platforms, or creating real-time quote generation via AI-driven workflow to automatically follow up with client services.
- Designed and implemented a scalable cloud-based solution on Microsoft Azure utilizing serverless App Services and Functions. Leveraged Azure SQL Database for structured data and Blob Storage for unstructured data. Possess strong knowledge of Oracle and SQL, along with familiarity with newer technologies like Dataverse, Databricks, and NoSQL.
- Developing Aged Care Regulator business process workflows using Power Automate as a low-code frontend, with Dataverse as the backend for data storage, and integrating recorded activities in Dynamics 365 and SharePoint 2019 Document List.
- This project streamlined Aged Care decision-makers' business processes by automating document approvals across various platforms (on-premises, cloud, SaaS).
- Designed and developed a document approval workflow automation system using C# (.NET) or Java. The system can potentially integrate with existing Oracle databases used by the legal team for data retrieval or population.
- Utilized Azure SQL Database for centralized storage and management of workflow data specific to the automation process.
- Integrated Power Automate to trigger and manage automated workflows based on document uploads within the Document Management System (DMS).
- Developed a serverless document processing system using Azure Functions to trigger Power Automate workflows upon document uploads in Azure Blob Storage, streamlining decision-makers' document review and approval processes.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, with a focus on M365 environments (O365, MS Teams, SharePoint 2019, Dynamics 365 CRM, Power BI, Power Automate, Power Apps etc.).
- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics 365 CRM, Power BI, Power Platform and SharePoint 2019 online
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems
- Migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL Database. Utilized Power BI Pro Get Data connectors for initial exploration and data access, followed by the Azure Database Migration Service for comprehensive database migration.
- Provide Level 2/3 technical and systems support for Microsoft infrastructures on Azure (PaaS / IaaS), covering AAD, DNS, DHCP, Exchange 2019, Windows Server 2022, and Microsoft 365 applications such as Office 365, Power BI, MS Teams, SharePoint 2019, Power Apps, Power Automate and Dynamics 365 CRM online, including Networking and Monitoring.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Deployed multi-cloud databases: Utilized either Azure Cosmos DB or Azure Blob Storage for storing unstructured vector data, such as social media posts, videos, images, and audio.
- Implemented Vector Search: Facilitated efficient retrieval from vector databases using native retrieval methods with Azure Cognitive Search, libraries like Faiss or Milvus (compatible with Azure), and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintex K2 Five workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- Migrated on-premise SharePoint 2019 with Nintex K2 forms/workflows to Azure SharePoint online and upgraded to Nintex K2 Five workflows.

- Performing Nintex K2 administrative tasks using K2 Configuration Analysis, K2 Designer, K2 Package and Deployment, K2 Server (Service account), K2 Management, Managing application administrators list, Workflow task approvers lists etc.
- **Automated Salesforce Development Workflow**:
- Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr IT Developer (Java/.Net /D365/MS Flow)/Snr DevOps Engineer/Python/AI/ML/DL/SP/BI Developer
Responsibilities:

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure Logic Apps (or specific Azure IPaaS service), BizTalk workflow, Microsoft Flow, SharePoint 2016, and Dynamics CRM 2016.

Designed and implemented a scalable cloud-based solution on Microsoft Azure utilizing serverless App Services and Functions. Leveraged Azure SQL Database for structured data and Blob Storage for unstructured data. Possess strong knowledge of Oracle and SQL, along with familiarity with newer technologies like Dataverse, Databricks, and NoSQL.

Designed and developed a document approval workflow automation system using C# (.NET) or Java. The system can potentially integrate with existing Oracle databases used by the legal team for data retrieval or population.

Utilized Azure SQL Database for centralized storage and management of workflow data specific to the automation process.

Integrated Power Automate to trigger and manage automated workflows based on document uploads within the Document Management System (DMS).

Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with RBA's enterprise services layer, databases, and host systems.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Research, design, development, testing, and maintenance of innovative and engaging digital solutions that support the firm's operations and client services by developing Dynamics 365/Power Automate/Salesforce/Mulesoft workflows.

Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.

Automated Salesforce Development Workflow: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for streamlined development and deployment. The pipeline utilizes `dfx force:source:pull` command to retrieve source code from a Salesforce org (e.g., `mydevorg`) using a specific username (`myuser@mydomain.com`).

Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.

Implemented automated security scans (SAST, DAST) into the GitLab CI pipeline, enforced granular access controls using roles, and managed secrets securely. Regular audits ensured compliance with industry standards like PCI-DSS or SOC 2.

- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.

Automation and CI/CD: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Used MuleSoft Anypoint Platform (including API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.

Data Pipeline Automation: Built and orchestrated data pipelines with Azure Data Factory (for orchestration), Azure Databricks, and Azure Event Hubs. This enabled efficient data movement and transformation. Normalized data in Azure Cosmos DB and Azure SQL Database for optimal performance.

Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

Data Transformation Expertise: Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.

Design and implement secure and scalable CI/CD pipelines using Azure DevOps, Azure SQL Database & Azure Blob Storage with Bitbucket repositories. Maintain DevOps Maturity Plan, Standardised Process across squads.

Optimized CI/CD pipelines with DevOps tools: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.

Production-Ready Machine Learning Models: Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.

CI/CD and DevSecOps Automation: Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Integration Solution Architect /Snr IT Developer (SharePoint/K2 Workflow)/Snr DevOps Engineer/DBA
Responsibilities:

Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.

Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.

Streamlined International Student Workflows: Develop custom applications on the Salesforce platform (Sales Cloud, Service Cloud, etc.) tailored for managing international student needs at the University of Canberra. These applications could include functionalities like:

10. Online application portal for international students: Visa and immigration document tracking;
11. Accommodation and orientation program management; Course enrollment and academic progress monitoring
12. These applications can leverage Salesforce DX for efficient development and deployment.

Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of student data.

Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio

Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc

Designed and maintained robust CI/CD pipelines using platform-specific tools:

13. **Azure: Azure DevOps**
14. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
15. **GCP: Cloud Build/Cloud Deploy**

Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:

Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or

Building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.

Developed and deployed secure, RESTful Python APIs for web applications using frameworks like Flask or FastAPI. These APIs can integrate with MLOps pipelines for data access, model predictions, or other machine learning tasks.

Applied Async programming using Python Asyncio to optimize system performance.

Leveraged frameworks like Flask and Django to build robust and scalable components.

Implemented API endpoints for data exchange using Python, Django, Flask and FastAPI

Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2016 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.

Managing, supporting, and troubleshooting MuleSoft, Azure IPaaS, BizTalk workflow, SharePoint 2016, Dynamics CRM 2016, and SQL Server 2016 on Windows Server 2016, with a strong focus on system and procedure documentation.

Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk workflow, SharePoint, and Dynamics CRM 2016.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.

Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).

Developed and deployed MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**

Healthcare Sector: IT Developer (SharePoint/Nintex Workflow)/DevOps Engineer /Software Test Engineer/Linux Admin.
Responsibilities:

- Solution and Feature design and implementation across multiple Web Applications with EML Work Cover Assessment Nintex Workflow in SharePoint 2013 Online with Office 365 /Nintex2013/CRM2013/SQL Server 2012/SSRS platforms.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Provided technical support to end users, resolving issues related to permissions, access, and usability.
- Contributed to the development of documentation and training materials for SharePoint users.
- Expand Dynamics CRM2013 functionality by customising Dynamics CRM to better fit the company and customers' needs
- Developing Dynamics CRM plugins either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Developing custom farm solutions in SharePoint 2013 using Design Manager, SP2013 Designer and TFS2015/VS2015
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD, Australia
Government Council: IT Developer (SharePoint /Nintex Workflow)/MS Dynamics CRM Developer
Responsibilities:

- Solution Architect to provide SharePoint /Nintex Solution designs and technical architecture for SCC SharePoint 2010 farm with F5 application load balancing in front-end web servers and Clustering Failover for back-end SQL database servers.
- Gathering requirements from stakeholders and translate them into technical solution design for SharePoint/CRM online.
- Consulting and development on BPM products like K2, Nintex, Biztalk, Oracle BPM Suite11gR2, Pega BPM, IBM TM1 etc
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Perform technical business analytical work and suggest how to improve AS-IS process and develop TO-BE process enhancements through business operation automation by implementing SharePoint /Nintex workflow solutions.
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Determing the record type of managed SharePoint content and how SharePoint Metadata is to be captured (Mapping columns to HP TRIM fields, Configuring Custom Mappings, maintenance of mappings and mapped values)
- Apply records management to SharePoint Content; Records management options(RMO);Enabling the use of RMOs, Site Level RMOs, List level RMOs and Self Service BI with SharePoint 2010
- Using SharePoint 2010 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or PRC Call(owssrc.dll), SPServices, ASMX Web Services, Rest API's: http://spserver/_vti_bin/listdata.svc
- Using C#, MVC5, App for SharePoint in VS2010 to deliver SharePoint 2010 /Nintex solutions in an enterprise environment.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- Integrated Yammer with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.
- Developing web parts, event receivers, web portals, web service via VS2010TFS,CAML in SharePoint 2010 platform

Aug.2009 - Oct. 2011 Manulife Financial Corporation (MLJ), Tokyo
Finance/Insurance Domain: DevOps Engineer /Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- DevOps deployment across multiple cloud platforms (AWS, Salesforce, Windows Azure, Oracle Exa Cloud).
- Install and configure CAS Server interfacing centric Active Directory to setup Exchange 2010 HA Network Load Balancing
- Create and develop Content Query Webpart with XSLT, Search Results Webpart for Enterprise Search Center.

- Branding SharePoint2010 by Yeoman, SharePoint Designer2010 and deploy MegaMenu for corp landing pages in VS2010.
- Design and developing Dynamic Data Entry User Interfaces, ASP.Net Dynamic Data Scaffolding and Page Templates.
- Performance Tuning of Applications via SSIS on BIDS2008R2 platform with capacity to produce high level technical documentations and results via SSRS; Develop, monitor and review migration of ETL packages using SSIS.
- Integrate AJAX into a Spring MVC application using JQuery on a client side for sending websphere JSM queries.
- Create financial reports either by SSRS or SAP Crystal Reports for VS2011 or Business Object BOXIR2 on HP-UX platform
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Westpac – DevOps Engineer – Lanson Sarah – 10/6/2024 Public Holiday – Not selected in 2 days.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Java/Python Developer/Snr Data Engineer/Integration Solution Architect/Snr Systems Engineer/ Salesforce/Mulesoft/AI/ML/DL/BI/SharePoint /D365 CRM Developer /EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Java/Python /AI/ML/DL/BI/SharePoint/D365 CRM Developer /Snr Data Engineer /EDW
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Design and develop software solutions, ensuring adherence to relevant standards and processes.
- Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with ACQSC's enterprise services layer, databases, and host systems.
- Full-stack engineer skilled in React/TypeScript for frontend development, Java/APIs for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and J2EE to ensure seamless data flow using RESTful web services.
- Collaborate closely with development and QA teams to ensure efficient application development and deployment.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation templates or Terraform. Utilized AWS Control Tower for centralized management of multiple AWS accounts, enabling parallel processing for various projects.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Design and implement secure and scalable CI/CD pipelines using AWS CodeBuild, RDS & S3 with Bitbucket repositories.
- Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Utilized Terraform's merge function to consolidate configuration data across various maps and objects, promoting efficient and maintainable infrastructure code. Experienced in Agile and able to promote Agile practices across IT teams.
- **Ensured high-quality, secure code**: Utilized SonarQube for code analysis, JFrog and Artifactory for artifact management, and Fortify Scan for security vulnerability detection, throughout the development process.
- **Integration Solution Design with AWS CodeBuild (CI/CD)**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Lambda functions, or Hybrid).
- Leverage AWS CodePipeline and AWS CodeDeploy to identify bottlenecks and inefficiencies in the CI/CD pipeline, and proactively implement improvements with AWS Lambda functions.
- **Supercharge automation!** Combine custom logic in Python or Go with Workato's pre-built integrations to streamline even the most complex business workflows. Maintain DevOps Maturity Plan, Standardised Process across squads.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.

- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer/Mulesoft/Java/Python/R/AI/ML/DL/BI/SharePoint Developer /Snr Data Engineer
Responsibilities:

Hybrid Cloud Architect: Established secure connections between AWS, GCP, and Azure cloud environments for data transfer. Utilized infrastructure-as-code tools (Terraform/Ansible, or Puppet) for automated and consistent provisioning.

Built secure and scalable Azure Landing Zones using Infrastructure as Code (IaC) with Azure Blueprints, ARM templates, or Terraform. Utilized Azure Lighthouse for centralized governance and management of multiple Azure subscriptions, enabling efficient parallel processing for various projects.

Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with RBA's enterprise services layer, databases, and host systems.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.

Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.

Implemented automated security scans (SAST, DAST) into the GitLab CI pipeline, enforced granular access controls using roles, and managed secrets securely. Regular audits ensured compliance with industry standards like PCI-DSS or SOC 2.

- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.

Automation and CI/CD: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Streamlined Terraform configuration management through the strategic use of the merge function to consolidate settings from diverse sources.

Used MuleSoft Anypoint Platform (including API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.

Data Pipeline Automation: Built and orchestrated data pipelines with Azure Data Factory, Azure Event Hubs, Azure Data Factory (Orchestration) for efficient data movement & transformation. Normalized data in Azure Cosmos DB & Azure SQL Database for optimal performance.

Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.

CI/CD for Integration Solutions: Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).

Data Transformation Expertise: Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.

Design and implement secure and scalable CI/CD pipelines using Azure DevOps, Azure SQL Database & Azure Blob Storage with Bitbucket repositories. Maintain DevOps Maturity Plan, Standardised Process across squads.

Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.

Optimized CI/CD pipelines with DevOps tools: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.

Production-Ready Machine Learning Models: Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.

CI/CD and DevSecOps Automation: Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
AI-Powered Pattern Recognition: Used Artificial Neural Networks to model complex patterns and forecast potential issues.
High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.
Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Integration Solution Architect/Snr DevOps Engineer /Java/Salesforce/Mulesoft/SharePoint /K2 Developer
Responsibilities:
Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.
Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
Designed and maintained robust CI/CD pipelines using platform-specific tools:

16. **Azure: Azure DevOps**
17. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
18. **GCP: Cloud Build/Cloud Deploy**

Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.
Developed and deployed secure, RESTful Python APIs for web applications using frameworks like Flask or FastAPI. These APIs can integrate with MLOps pipelines for data access, model predictions, or other machine learning tasks.
Applied Async programming using Python Asyncio to optimize system performance.
Leveraged frameworks like Flask and Django to build robust and scalable components.
Implemented API endpoints for data exchange using Python, Django, Flask and FastAPI
Collaborate closely with development and QA teams to ensure efficient application development and deployment.
Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.
Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.
Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.
Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.
Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
Developed and deployed a MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.
Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of student data.
CI/CD and DevSecOps Automation (AWS): Built and maintained CI/CD pipelines for various applications, employing DevSecOps practices using industry-standard tools (CodeBuild/GitHub) & following best practices for secure development

- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Data Transformation Expertise (Azure):** Employed Azure Logic Apps (Connectors, Logic Flows) for complex data transformations, leveraging Azure Data Factory (ADF) for standardization and basic transformations.

- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or on-premises environment.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Automation:** Created and maintained automated scripts using Python or Perl programming, along with Bash or Shell scripting, to streamline development and operational processes.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- **Collaboration:** Work closely with software developers, QA engineers, and system administrators to ensure smooth deployment and operation of applications.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- **Troubleshooting:** Diagnose and resolve technical issues related to CI/CD pipelines, containerization, and infrastructure.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Design and develop cloud-native applications using AWS services and Python
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).

- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code,

Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior DevOps & Security Analyst – DCS – Syd- 29/5/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps & Security Analyst/Snr Data Engineer/Integration Solution Architect/Snr Systems Engineer/Snr Linux Admin. /Python/R /Salesforce/Mulesoft/AI/ML/DL/BI/SharePoint /D365 CRM Developer /EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr DevOps & Security Analyst/Snr Data Engineer/Python/R/AI/ML/DL/BI/SPoint/D365 CRM Developer/EDW
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Lead the implementation of automated software components and system changes to achieve service standards.
- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer:** Utilize Amazon EMR, a scalable managed Hadoop framework service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Use Amazon Kinesis Data Streams as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the Silver & potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Unleash automation magic! Combine code (Python, Go) for custom logic with Workato's pre-built integrations to seamlessly automate complex workflows across your business, simplifying even the most intricate processes.
- Crafted AI-driven pipelines in Python to automate GCP IaC pipeline creation, prediction and optimization using Cloud Build, Cloud Deployment Manager and Cloud SDK: client = googleapiclient.discovery.build('deploymentmanager', 'v2')...
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Eextensive hands-on experience managing and supporting: Intrusion and detection systems (IDS), firewalls, Content Data Network (CDN), encryption, code analysis, Cyber Security incident and response, identity and access management (IAM)
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- Secure secrets with HashiCorp Vault or AWS Secrets Manager. Implement network segmentation and firewalls.
- Integrated security scanning into CI/CD pipelines to shift security left and prevent vulnerabilities from reaching production.
- **Secured API Access for Aged Care Patients' Records**: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the aged care patients' records system and external partners. This ensured the privacy and security of sensitive patient data.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
  - **Azure: Azure DevOps**
  - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**

- **GCP: Cloud Build/Cloud Deploy**

- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to your specific needs. Or
- **Java with Spring Batch along with Kafka and Spark** to provide powerful tools for managing complex data pipelines within your CI/CD process.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Cloud Security with AI**: Leveraged AI tools to detect and prevent threats, manage vulnerabilities, and secure data. Such as McAfee ESM and Palo Alto Networks Cortex XDR for impact.
- **Network Security via SIEM:** Hybrid Cloud Security with Netskope SASE: Deploy Netskope Service Edge in data centers and cloud environments. Configure SASE functionalities including CASB, SWG, ZTNA, and DLP. Establish secure connections between on-premises and cloud (e.g., ExpressRoute, Direct Connect).

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps & Security Analyst/Snr Data Engineer/Python/R/AI/ML/DL/BI/SharePoint Developer /EDW
Responsibilities:

Hybrid Cloud Architect: Established secure connections between AWS, GCP, and Azure cloud environments for data transfer. Utilized infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning.

Built and implemented Medallion Architecture with Data Lake in GCP:

Batch Layer: Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, & unrefined data.

Serving Layer: Implemented BigQuery with its built-in Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.

Real-time Layer: Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.

Configured a Netskope Cloud Private Access (NCPA) tenant within the Azure cloud environment and established secure connections between Azure and AWS environments to Netskope using either Azure ExpressRoute or AWS Direct Connect

Defined granular policies for user authentication methods (SAML for Azure AD, LDAP for AWS IAM) to grant secure access to designated resources. Enabled comprehensive security features like cloud malware protection, data loss prevention (DLP), and web filtering across both cloud environments, creating a robust hybrid cloud security posture.

Data Pipeline Automation: Built & orchestrated data pipelines with dbt, Cloud Dataflow, Cloud Pub/Sub, Cloud Composer for efficient data movement & transformation. Normalized data in Cloud Spanner & Cloud SQL for optimal performance.

Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

CI/CD for Integration Solutions: Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).

Cloud-agnostic security by using scripting languages (Python, Bash, and PowerShell) to integrate container scanning tools like Snyk and Aqua Security.

Data Transformation Expertise: Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.

ETL/API Orchestration (Platform Specific): Designed, developed, and maintained robust ETL pipelines/API orchestration using platform-specific tools like:

19. Azure: Synapse Analytics with Data Factory/Pipelines and Azure DevOps for data processing.
20. AWS: Glue for data processing and integration with AWS services for loading data into Aurora databases.
21. GCP: Cloud Dataflow/Data Fusion for data processing and BigQuery for data warehousing and analytics.

Customizable ETL/API Orchestration: Alternatively, developed custom ETL/API orchestrations using Python/Pandas with Airflow or Java/Spring Batch with Kafka and Spark for any cloud or on-premises environment.

Multi-layered API security for data integrity and protection: Protect sensitive data from unauthorized access and manipulation through comprehensive security measures including authentication, authorization, encryption, and input validation. Implement these measures using REST API gateways like Anypoint API Gateway, Azure API Management, or AWS API Gateway. For enhanced security, leverage Anypoint Platform's built-in capabilities for:
API Throttling and Rate Limiting: Control the volume of API requests to prevent denial-of-service attacks.
Client ID Enforcement: Ensure only authorized applications can access your APIs.
Advanced Threat Protection: Mitigate common API vulnerabilities like SQL injection and cross-site scripting (XSS) attacks.
Tokenization: Securely store sensitive data by replacing it with non-identifiable tokens.
Centralized Policy Management: Manage and enforce API security policies consistently across your APIs.
Detailed API Analytics: Gain insights into API usage patterns and identify potential security risks.
Additionally, leverage monitoring solutions like Azure Monitor/Log Analytics or AWS CloudWatch logs for deeper analysis and anomaly detection.
Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.
High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.
Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
CI/CD with BuildKite and GitHub: Performed code releases using CI/CD tools (BuildKite, GitHub). Defined CI/CD steps within the BuildKite pipeline for fetching code, running tests, building applications, and deploying to target environments.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Integration Solution Architect/Snr DevOps & Security Analyst/Mulesoft/ML/SharePoint /K2 Developer
Responsibilities:
Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications.
Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.
Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.
Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.
Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.
Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.
Enhancing Endpoint Security: Implement and optimize Microsoft Defender and Azure Arc to bolster our endpoint security measures. **Implementing Security Controls:** Develop and enhance security controls for both new and existing systems.
Providing Expert Advice: Guide internal teams and technical vendors on cyber security best practices and solutions.
Conducting Risk Assessments: Perform thorough security risk assessments on technical designs to identify vulnerabilities.
DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
Developed and deployed a MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.
Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of student data.
Serverless Data Management (GCP): Demonstrated expertise in managing and analyzing complex, high-volume, high-dimensional data within a serverless GCP Lambda architecture. Leveraged BigQuery and Cloud Dataflow for efficient data processing and analysis.

- **Cloud-Agnostic Data Platform Design:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **CI/CD for Integration Solutions (GCP):** Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (Cloud Functions or Hybrid).
- **Data Transformation Expertise (GCP):** Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations, leveraging DBT for standardization and basic transformations.
- **ETL/API Orchestration (Platform Specific):** Designed, developed, and maintained robust ETL pipelines/API orchestration using platform-specific tools:
- Integrated security scanning into CI/CD pipelines in AWS using Amazon Inspector in a .yaml file.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- GCP: Cloud Dataflow/Data Fusion for data processing and BigQuery for data warehousing and analytics.
- **Production-Ready Machine Learning (GCP):** Engineered and deployed production-ready ML models for real-world impact using Python and GCP MLOps tools.
- **CI/CD and DevSecOps Automation (GCP):** Developed and maintained CI/CD pipelines (Cloud Build/GitHub) for various applications, employing DevSecOps practices for improved security.
- **AI-Powered Pattern Recognition:** Used Artificial Neural Networks to model complex patterns and forecast potential issues.
- API Development with Python: Developed and deployed MLOps pipelines using Python frameworks (e.g., Flask, FastAPI) and implemented secure RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: DevOps & Security Analyst/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **CI/CD Pipeline Management:** Design, implement, and manage continuous integration and continuous deployment (CI/CD) pipelines using Jenkins.
- **Containerization:** Develop, deploy, and maintain containerized applications using Docker. Ensure applications are scalable and resilient.
- **Automation:** Create & maintain automated scripts using Shell scripting to streamline development & operational processes.
- Implements, maintains, and analyses security measures to protect the organisation's ICT infrastructure environment and data according to the information security policies and framework..
- **Monitoring and Logging:** Implement and maintain monitoring and logging solutions to ensure system reliability and performance.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Completed design documentation, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps & Security Analyst/Data Engineer /Software Test Engineer/SharePoint Consultant
Responsibilities:

- Design and develop cloud-native applications using AWS services and Python
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda
- Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- **Self-driven & Collaborative:** Demonstrate strong problem-solving skills, a collaborative spirit, and a commitment to staying current with data technologies in Google Cloud.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Snr Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior DevOps Engineer – EKS-GKE-TRC- 19/5/2024 – Maybe for Macquirie Bank – Anne-Maria closed on 28/5
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Snr Data Engineer/Integration Solution Architect/Snr Systems Engineer/Snr Linux Admin. /Python/R /Salesforce/Mulesoft/AI/ML/DL/BI/SharePoint /D365 CRM Developer /EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Principle DevOps Engineer/Snr Data Engineer/Python/R /AI/ML/DL/BI/SharePoint/D365 CRM Developer/EDW
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Design, implement & manage secure, scalable DevOps infrastructure to ensure reliable deployment and operation of the Commission's systems and applications. This included building and maintaining robust automated deployment pipelines.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- **Integration Solution Design with AWS CodeBuild (CI/CD)**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Lambda functions, or Hybrid).
- Used MuleSoft Anypoint Platform (including API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.
- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.

- GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Developed and maintained CI/CD pipelines for various applications using GCP/AWS/Azure DevOps, integrating DevSecOps practices and incorporating AIOps with MLOps for intelligent automation.
- Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Unleash automation magic! Combine code (Python, Go) for custom logic with Workato's pre-built integrations to seamlessly automate complex workflows across your business, simplifying even the most intricate processes.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer /Snr Data Engineer/Python/R/AI/ML/DL/BI/SharePoint Developer /EDW
Responsibilities:

Hybrid Cloud Architect: Established secure connections between AWS, GCP, and Azure cloud environments for data transfer. Utilized infrastructure-as-code tools (Terraform/Ansible, or Puppet) for automated and consistent provisioning.

Collaborate with development and operations teams to optimize application performance and reliability.

Built and implemented Medallion Architecture with Data Lake in GCP:

Batch Layer: Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, & unrefined data.

Serving Layer: Implemented BigQuery with its built-in Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.

Real-time Layer: Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Data Pipeline Automation: Built & orchestrated data pipelines with dbt, Cloud Dataflow, Cloud Pub/Sub, Cloud Composer for efficient data movement & transformation. Normalized data in Cloud Spanner & Cloud SQL for optimal performance.

Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

CI/CD for Integration Solutions: Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).

Data Transformation Expertise: Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.

ETL/API Orchestration (Platform Specific): Designed, developed, and maintained robust ETL pipelines/API orchestration using platform-specific tools like:

22. Azure: Synapse Analytics with Data Factory/Pipelines and Azure DevOps for data processing.
23. AWS: Glue for data processing and integration with AWS services for loading data into Aurora databases.
24. GCP: Cloud Dataflow/Data Fusion for data processing and BigQuery for data warehousing and analytics.

Customizable ETL/API Orchestration: Alternatively, developed custom ETL/API orchestrations using Python/Pandas with Airflow or Java/Spring Batch with Kafka and Spark for any cloud or on-premises environment.

Implemented GCP cloud-native design patterns, including leveraging Stackdriver Monitoring for comprehensive monitoring and observability of GCP resources. This includes tracking metrics, logs, and events from both applications & infrastructure. Additionally, utilized Cloud IAM for robust user identity and access management in web and mobile applications.

Production-Ready Machine Learning Models: Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.

CI/CD and DevSecOps Automation: Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.

AI-Powered Pattern Recognition: Used Artificial Neural Networks to model complex patterns and forecast potential issues.

High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.

Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.

Developed Python scripts that leverage AI to automate the creation, prediction, and optimization of GCP Infrastructure as Code (IaC) pipelines. This utilizes Cloud Build for building and running pipelines, Cloud Deployment Manager for managing infrastructure deployments, and the Cloud SDK for interacting with GCP services.

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Security and Compliance: **Implement security best practices**: Regularly patch systems and applications.

Secure secrets with HashiCorp Vault or AWS Secrets Manager. Implement network segmentation and firewalls.

Ensure compliance with industry standards & internal policies. Integrate security canning tools throughout development.

Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau

Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)

Gov. University: Integration Solution Architect/Snr DevOps Engineer /Salesforce/Mulesoft/ML/SharePoint /K2 Developer

Responsibilities:

Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications.

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.

Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.

Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.

Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.

Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.

Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).

Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).

Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).

Developed and deployed a MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.

Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. T

his ensured the privacy and security of student data.

Serverless Data Management (GCP): Demonstrated expertise in managing and analyzing complex, high-volume, high-dimensional data within a serverless GCP Lambda architecture. Leveraged BigQuery and Cloud Dataflow for efficient data processing and analysis.

- **Automated Data Pipelines (GCP):** Built and orchestrated efficient data pipelines using dbt, Cloud Dataflow, Cloud Pub/Sub, and Cloud Composer for data movement and transformation. Normalized complex data in Cloud Spanner and Cloud SQL for optimal performance within GCP.
- **Cloud-Agnostic Data Platform Design:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **CI/CD for Integration Solutions (GCP):** Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (Cloud Functions or Hybrid).
- **Data Transformation Expertise (GCP):** Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations, leveraging DBT for standardization and basic transformations.
- **ETL/API Orchestration (Platform Specific):** Designed, developed, and maintained robust ETL pipelines/API orchestration using platform-specific tools:
- GCP: Cloud Dataflow/Data Fusion for data processing and BigQuery for data warehousing and analytics.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or on-premises environment.
- **Production-Ready Machine Learning (GCP):** Engineered and deployed production-ready ML models for real-world impact using Python and GCP MLOps tools.
- **Multi-Cloud Databases for Unstructured Data (GCP):** Utilized Cloud Firestore for storing unstructured data (e.g., social media posts, videos).
- **CI/CD and DevSecOps Automation (GCP):** Developed and maintained CI/CD pipelines (Cloud Build/GitHub) for various applications, employing DevSecOps practices for improved security.
- **AI-Powered Pattern Recognition:** Used Artificial Neural Networks to model complex patterns and forecast potential issues.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- API Development with Python: Developed and deployed MLOps pipelines using Python frameworks (e.g., Flask, FastAPI) and implemented secure RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.
- Continuously evaluate and integrate new technologies to improve efficiency and scalability.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **CI/CD Pipeline Management:** Design, implement, and manage continuous integration and continuous deployment (CI/CD) pipelines using Jenkins.
- **Containerization:** Develop, deploy, and maintain containerized applications using Docker. Ensure applications are scalable and resilient.
- **Automation:** Create and maintain automated scripts using Shell scripting to streamline development and operational processes.
- **Monitoring and Logging:** Implement and maintain monitoring and logging solutions to ensure system reliability and performance.
- **Collaboration:** Work closely with software developers, QA engineers, and system administrators to ensure smooth deployment and operation of applications.
- **Security:** Implement security best practices for the CI/CD pipeline and containerized applications.
- **Troubleshooting:** Diagnose and resolve technical issues related to CI/CD pipelines, containerization, and infrastructure.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Completed design documentation, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.

- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Played a key role in container orchestration using Kubernetes or Nomad, enabling dynamic scaling of services.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SP Consultant
Responsibilities:

- Design and develop cloud-native applications using AWS services and Python
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Snr Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Actual is Data/DevOps Engineer – Correlated Resource – Bella Vista Woolworths – 19/5/2024- interviewed on 23/5/24
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Snr Data Engineer/Integration Solution Architect/Snr Systems Engineer/Snr Linux Admin. /Python/R /Salesforce/Mulesoft/AI/ML/DL/BI/SharePoint /D365 CRM Developer /EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Principle DevOps Engineer/Snr Data Engineer/Python/R /AI/ML/DL/BI/SharePoint/D365 CRM Developer/EDW
Responsibilities:

- **Lead CI/CD Pipeline for High-Volume Data Engineering:** Refine and manage CI/CD pipelines for processing intricate, high-volume, high-dimensional data from various sources within the Cyber Data Engineering team.
- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- **Architected and implemented a modern 3-layer lambda architecture on GCP**:
- **Batch layer**: Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed BigQuery with its built-in Spark capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the data warehouse for advanced analytics and reporting.
- Built & orchestrate data pipelines with dbt, Cloud Dataflow, Cloud Pub/Sub, & Cloud Composer for efficient data movement and transformation. Then, normalized complex data in Cloud Spanner and Cloud SQL for optimal performance within GCP.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- **Integration Solution Design with GCP Cloud Build (CI/CD)**: **Design Integration Flow**:
- Define data flow, transformations, and choose the orchestration platform (MuleSoft, Cloud Functions, or Hybrid).

- Used MuleSoft Anypoint Platform (including API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.
- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Deployed multi-cloud databases**: Utilized either Cloud Spanner or Cloud Firestore for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Faiss, Firestore Extensions, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed and maintained CI/CD pipelines for various applications using GCP/AWS/Azure DevOps, integrating DevSecOps practices and incorporating AIOps with MLOps for intelligent automation.
- Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.

Automating Machine Learning Workflows on GCP: Utilize GCP AI Platform or Vertex AI to automate the machine learning development process. Similar to an automated factory where robots build other robots driven by its pre-built components and collaboration tools to streamline and optimize AI development through MLOps capabilities.

- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Unleash automation magic! Combine code (Python, Go) for custom logic with Workato's pre-built integrations to seamlessly automate complex workflows across your business, simplifying even the most intricate processes.
- Crafted AI-driven pipelines in Python to automate GCP IaC pipeline creation, prediction and optimization using Cloud Build, Cloud Deployment Manager and Cloud SDK: client = googleapiclient.discovery.build('deploymentmanager', 'v2')...

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer /Snr Data Engineer/Python/R/AI/ML/DL/BI/SharePoint Developer /EDW
Responsibilities:

Hybrid Cloud Architect: Established secure connections between AWS, GCP, and Azure cloud environments for data transfer. Utilized infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning.

Built and implemented Medallion Architecture with Data Lake in GCP:

Batch Layer: Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, & unrefined data.

Serving Layer: Implemented BigQuery with its built-in Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.

Real-time Layer: Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Data Pipeline Automation: Built & orchestrated data pipelines with dbt, Cloud Dataflow, Cloud Pub/Sub, Cloud Composer for efficient data movement & transformation. Normalized data in Cloud Spanner & Cloud SQL for optimal performance.

Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

CI/CD for Integration Solutions: Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).

Data Transformation Expertise: Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.

ETL/API Orchestration (Platform Specific): Designed, developed, and maintained robust ETL pipelines/API orchestration using platform-specific tools like:

25. Azure: Synapse Analytics with Data Factory/Pipelines and Azure DevOps for data processing.
26. AWS: Glue for data processing and integration with AWS services for loading data into Aurora databases.

27. GCP: Cloud Dataflow/Data Fusion for data processing and BigQuery for data warehousing and analytics.

Customizable ETL/API Orchestration: Alternatively, developed custom ETL/API orchestrations using Python/Pandas with Airflow or Java/Spring Batch with Kafka and Spark for any cloud or on-premises environment.

Production-Ready Machine Learning Models: Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.

Multi-Cloud Databases for Unstructured Data (GCP): Utilized Cloud Firestore for storing unstructured data (e.g., social media posts, videos).

CI/CD and DevSecOps Automation: Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.

AI-Powered Pattern Recognition: Used Artificial Neural Networks to model complex patterns and forecast potential issues.

High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.

Machine Learning Automation (GCP): Performed Machine Building the Machine (MBM) tasks on GCP AI Platform using pre-built components and collaboration tools for streamlined and optimized AI development.

Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.

CI/CD with BuildKite and GitHub: Performed code releases using CI/CD tools (BuildKite, GitHub). Defined CI/CD steps within the BuildKite pipeline for fetching code, running tests, building applications, and deploying to target environments.

Developed Python scripts that leverage AI to automate the creation, prediction, and optimization of GCP Infrastructure as Code (IaC) pipelines. This utilizes Cloud Build for building and running pipelines, Cloud Deployment Manager for managing infrastructure deployments, and the Cloud SDK for interacting with GCP services.

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile team environment, DevOps and Configuration-As-Code)

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau

Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Integration Solution Architect/Snr DevOps Engineer /Salesforce/Mulesoft/ML/SharePoint /K2 Developer
Responsibilities:

Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications.

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.

Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.

Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.

Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.

Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.

Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.

Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).

Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).

Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).

Developed and deployed a MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.

Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. T
his ensured the privacy and security of student data.

Serverless Data Management (GCP): Demonstrated expertise in managing and analyzing complex, high-volume, high-dimensional data within a serverless GCP Lambda architecture. Leveraged BigQuery and Cloud Dataflow for efficient data processing and analysis.

- **Automated Data Pipelines (GCP):** Built and orchestrated efficient data pipelines using dbt, Cloud Dataflow, Cloud Pub/Sub, and Cloud Composer for data movement and transformation. Normalized complex data in Cloud Spanner and Cloud SQL for optimal performance within GCP.
- **Cloud-Agnostic Data Platform Design:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **CI/CD for Integration Solutions (GCP):** Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (Cloud Functions or Hybrid).
- **Data Transformation Expertise (GCP):** Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations, leveraging DBT for standardization and basic transformations.
- **ETL/API Orchestration (Platform Specific):** Designed, developed, and maintained robust ETL pipelines/API orchestration using platform-specific tools:
- GCP: Cloud Dataflow/Data Fusion for data processing and BigQuery for data warehousing and analytics.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or on-premises environment.
- **Production-Ready Machine Learning (GCP):** Engineered and deployed production-ready ML models for real-world impact using Python and GCP MLOps tools.
- **Multi-Cloud Databases for Unstructured Data (GCP):** Utilized Cloud Firestore for storing unstructured data (e.g., social media posts, videos).
- **CI/CD and DevSecOps Automation (GCP):** Developed and maintained CI/CD pipelines (Cloud Build/GitHub) for various applications, employing DevSecOps practices for improved security.
- **AI-Powered Pattern Recognition:** Used Artificial Neural Networks to model complex patterns and forecast potential issues.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- API Development with Python: Developed and deployed MLOps pipelines using Python frameworks (e.g., Flask, FastAPI) and implemented secure RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **CI/CD Pipeline Management:** Design, implement, and manage continuous integration and continuous deployment (CI/CD) pipelines using Jenkins.
- **Containerization:** Develop, deploy, and maintain containerized applications using Docker. Ensure applications are scalable and resilient.
- **Automation:** Create and maintain automated scripts using Shell scripting to streamline development and operational processes.
- **Monitoring and Logging:** Implement and maintain monitoring and logging solutions to ensure system reliability and performance.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Completed design documentation, data models, loading approaches and source to target mapping documents.

- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Played a key role in container orchestration using Kubernetes or Nomad, enabling dynamic scaling of services.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SP Consultant
Responsibilities:

- Design and develop cloud-native applications using AWS services and Python
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda
- Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- **Self-driven & Collaborative:** Demonstrate strong problem-solving skills, a collaborative spirit, and a commitment to staying current with data technologies in Google Cloud.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Snr Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.

- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer – Salesforce-Mulesoft - NDIA- Recruiment Hive – 13/5/2023
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Snr Integration Developer/Snr Data Engineer/Java/.Net Developer/Clouds/Linux Administrator Salesforce /MuleSoft Developer /AI/ML/BI/SharePoint/D365 CRM Devloper/EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Principle DevOps Engineer/Snr Integration Developer/Java/.Net /Python/R /AI/ML/SP/D365 CRM Developer
Responsibilities:

- Lead the design, implementation, and support of CICD automation solutions and processes to support Release efficiency and quality in ACQSC Salesforce and MuleSoft platforms
- **Leveraged MuleSoft Anypoint Platform for Hybrid Cloud Integration**: Designed APIs using MuleSoft's hybrid capabilities to seamlessly integrate between existing AWS infrastructure (API Gateway) and Azure IPaaS environments.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintet K2 Five workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- **Automated Salesforce Development Workflow**:
- Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.
- Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.

- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx force:source:deploy`. (Optional: additional deployment steps like code analysis and MuleSoft application deployments if applicable)
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- **Automated Deployments with SFDX:** Minimized manual errors and accelerated releases.
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
- **Streamlined Change Management**: Utilized Salesforce change sets (or a version control system like Git, GitLab, Azure Repos, or AWS CodeCommit) within the CICD flow for managing code changes.
- **Automated Metadata**: Streamlined metadata retrieval & deployment for efficient development and release cycles.
- **MuleSoft Integration**: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
- **Integration Solution Design with AWS CodePipeline**: Design Integration Flow: Define data flow, transformations, and choose the orchestration platform (MuleSoft, AWS Step Functions, or Hybrid).
- **Develop and Build**: Use AWS CodeBuild project with source code management (e.g., Git, AWS CodeCommit) to trigger builds for each integration platform (MuleSoft, AWS Step Functions, BizTalk).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, AWS CloudFormation templates for Step Functions/Cloud Integration).
- **Streamlined NDIA/Health Workflows**: Develop custom applications on the Salesforce platform (Sales Cloud, Service Cloud, etc.) tailored for the NDIA or health sector (e.g., **NDIA**: participant management, plan tracking, provider claims processing; **Health**: patient appointment scheduling, referral management, claims adjudication). These applications can leverage Salesforce DX for efficient development and deployment.
- **Enhanced User Experiences with Vlocity and OmniStudio**: Build low-code/no-code user interfaces and streamline processes within the Salesforce CRM using Vlocity and OmniStudio. This allows for rapid development of custom portals, forms, and mobile experiences for NDIA participants/providers (NDIA) or patients/healthcare providers (Health).
- **Data-Driven Decision Making with Powerful AI in CRM Analytics**: Integrate CRM Analytics (formerly Tableau CRM) to transform Salesforce data into actionable insights using its most powerful AI features. Public sector agencies can leverage capabilities like:
    - **Predictive Analytics**: Forecast future trends and potential issues related to participant needs, service utilization, or provider performance (NDIA) or patient demographics, treatment outcomes, resource allocation (Health).
    - **Einstein Discovery**: Automatically uncover hidden patterns and relationships within Salesforce data, leading to more informed recommendations and decisions for improved service delivery or resource management.
    - **Einstein Einstein Next Best Action**: Leverage AI to recommend the most effective next steps for NDIA staff (e.g., suggesting the most suitable plan for a participant) or healthcare providers (e.g., recommending the most appropriate treatment based on patient history).
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior DevOps Engineer/Snr Integration Developer/Snr Data Engineer/Java/AI/ML/ML/SP/BI Developer
Responsibilities:

- Led the design, implementation, and support of CICD automation solutions and processes to support Release efficiency and quality in RBA Salesforce and MuleSoft platforms

Streamlined Data Integration with MuleSoft: Utilized MuleSoft Anypoint Platform to design APIs that seamlessly integrate various data sources within the AWS cloud. This included leveraging pre-built connectors from Anypoint Exchange for services like AWS API Gateway, SQS, DynamoDB, & other relevant AWS services. The "write-once, deploy-anywhere" approach offered efficient data movement and simplified complex infrastructure management within the AWS ecosystem.

Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintxt workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
Automated Salesforce Development Workflow: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for streamlined development and deployment. The pipeline utilizes `dfx force:source:pull` command to retrieve source code from a Salesforce org (e.g., `mydevorg`) using a specific username (`myuser@mydomain.com`).
Integration Solution Design with AWS CodePipeline: Design Integration Flow: Define data flow, transformations, and choose the orchestration platform (MuleSoft, AWS Step Functions, or Hybrid).
Develop and Build: Use AWS CodeBuild project with source code management (e.g., Git, AWS CodeCommit) to trigger builds for each integration platform (MuleSoft, AWS Integration Services, BizTalk).
Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, AWS CloudFormation templates for Step Functions/Integration Services).
Automated Salesforce Development Workflow:
Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.
Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx force:source:deploy`. (Optional: additional deployment steps like code analysis and MuleSoft application deployments if applicable)
Streamlined Salesforce Development and Deployment: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
Automated Deployments with SFDX: Minimized manual errors and accelerated releases.
Automated Metadata: Streamlined metadata retrieval & deployment for efficient development and release cycles.
MuleSoft Integration: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
Design, develop, and maintain a robust ETL pipeline/API orchestration using either:

28. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
29. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
30. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
Performing data transformation and integration tasks either using Python in AWS CodeBuild to leverage the power of AWS CodePipeline to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Worked on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn
Integrated DevSecOps practices with Infrastructure as Code (IaC) security principles to automate security configurations and ensure consistent security posture across deployments.
Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Snr DevOps Engineer/Integration Developer/Snr Data Engineer/Python/ML/SharePoint/K2 Developer
Responsibilities:

- Led the design, implementation, and support of CICD automation solutions and processes to support Release efficiency and quality in UC Salesforce and MuleSoft platforms
- **Leveraged MuleSoft Anypoint Platform for Hybrid Cloud Integration**: Designed APIs using MuleSoft's hybrid capabilities to seamlessly integrate between existing AWS infrastructure (including API Gateway) and Microsoft Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminated the need for Terraform or complex infrastructure management, enabling greater agility and flexibility within the hybrid cloud environment
- **Streamlined International Student Workflows**: Develop custom applications on the Salesforce platform (Sales Cloud, Service Cloud, etc.) tailored for managing international student needs at the University of Canberra. These applications could include functionalities like:
    - Online application portal for international students; Visa and immigration document tracking;
    - Accommodation and orientation program management; Course enrollment and academic progress monitoring
- These applications can leverage Salesforce DX for efficient development and deployment.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- **Enhanced User Experiences with OmniStudio**: Build low-code/no-code user interfaces and streamline processes within the Salesforce CRM using OmniStudio. This allows for rapid development of custom portals, forms, and mobile experiences for international students at the University of Canberra. Examples include:
    - Self-service portal for international students to access resources, submit requests, and track applications;
    - Interactive mobile app for international students to access course information, campus maps, and connect with support services; Personalized communication channels to keep international students informed and engaged

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
Developing, customizing, and implementing data integration solutions for MS Dynamics 365 CRM using KingswaySoft SSIS Integration Toolkit for Microsoft Dynamics 365. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.
Managing, supporting, and troubleshooting MuleSoft, Azure IPaaS, BizTalk, SharePoint 2019, Dynamics 365 CRM, and SQL Server 2022 on Windows Server 2022, with a strong focus on system and procedure documentation.
Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless failover between AWS and Azure. This ensures resilience against outages.
Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
Enhanced Code Quality: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
Streamlined Change Management: Utilized Salesforce change sets (or a version control system like Git, GitLab, Azure Repos, or AWS CodeCommit) within the CICD flow for managing code changes.
Optimized code for ML projects using data structures, algorithms, and software architecture knowledge.
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
Applied natural language processing techniques such as text preprocessing, sentiment analysis and topic modeling using NLTK (Natural Language Toolkit) and Gensim.
Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer /Integration Developer /Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency
- Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases

- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Systems Integration Developer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Amazon CloudWatch for logging, partitioning, indexing, & optimizing AWS Data Pipeline structure to analyze system & application performance, providing insights for business case development to enhance service delivery & reduce issues.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Managed Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.

- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
Reporting: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
Clouds: MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
CMS: SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010~Nintex K2 Five Workflow and SharePoint 2007 ~ 2022 Workflows.

ETL Developer – Charterhouse – 8/May/2024 maybe for SIRA
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Snr ETL Developer/Principle Data Engineer/Senior DevOps Engineer/Software Engineer/Snr Linux Admin./Py /R Developer /Salesforce/Mulesoft /AI/ML/DL/BI/SharePoint/Dynamics 365 CRM Developer /EDW/DBA –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior ETL Developer/Principle Data Engineer/Snr DevOps Engineer/Python/R/AI/ML/SP/D365 CRM Developer
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Collaborate with business and technical teams to design, test, and analyze ETL processes, while managing architecture tasks like reverse engineering, unit testing, and data warehousing integration.
- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer:** Utilize Amazon EMR, a scalable managed Hadoop framework service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Manage the Commission's analytics environment, encompassing data ingestion, quality & modeling, ETL/API orchestration, visualization, metrics & reporting, and advanced research, all within a strict data governance framework.
- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Data development, optimization, and manipulation using PySpark libraries (PySpark Core, PySpark SQL, PySpark MLlib etc.) or SQL, including PySpark scripting.
- Saving a PySpark dataframe as a Delta table in a Fabric Warehouse (Lakehouse) and retrieving the same table in Metallion Warehouse (Data Lake or Delta Lake) in Auzre using spark.read.format("delta").load("<path_to_delta_table>")
- Developed ETL/ELT workflows for Azure Synapse Analytics, utilizing SSIS, SQL PolyBase on SQL Server 2022, Stream Analytics, and PowerShell scripting for optimized data movement and transformation.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Apply nested row context within calculations and conditional formatting with gradients.
- Owning the technical aspects of data and data pipelines, ensuring they adhere to data standards, architectural guidelines, and meet business requirements.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Snr ETL Developer/Principle Data Engineer/Snr DevOps Engineer/Python/R/Java/AI/ML/SP/BI Developer
Responsibilities:

Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and implemented a scalable Snowflake EDW (Terraform) for the hybrid Azure and AWS environment. Data from various sources (including on-premises databases) is ingested using Azure Data Factory (ADF). SSIS and Spark Notebooks handle data transformations, with Spark Notebooks leveraging Apache Spark for complex big data scenarios. This unified platform integrates with Machine Learning tools from both clouds, enabling advanced analytics and large-scale ML.
Manage the Commission's analytics environment, encompassing data ingestion, quality & modeling, ETL/API orchestration, visualization, metrics & reporting, and advanced research, all within a strict data governance framework.
Building data pipelines using Amazon EMR with PySpark to extract data from various sources (Oracle/Excel), store it in a Data Lake, and then transform and model it for analytics and reporting purposes.
Expert in data engineering and development using Databricks (Databricks Runtime, Apache Spark) on Microsoft Azure, specializing in hands-on ETL and Data Integration tasks.
Extract data from live websites into JSON format using Power BI, then utilize pandas and scikit-learn in Python for data classification and training within Jupyter Notebook. Integrate the results into Power BI dashboards or notebooks for trend prediction and visualization.
Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
Designed and implemented IaC solutions using Terraform as the primary IaC tool. Possess experience with defining infrastructure foundations for systems or Azure products using tools like ADF, ADB, ARM templates or LAB3 Bedrock, a pre-built IaC solution leveraging Terraform for faster cloud adoption based on the MS Cloud Adoption Framework (CAF).
Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through data modeling and statistical analysis.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Utilized SPARX for data modeling to support efficient pipeline development within a comprehensive enterprise architecture framework. This data model can then be used for integration with Snowflake, AWS, SAS, and Python in either a Salesforce or Mulesoft development environment.
Deigned and implemented data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.
Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.
Utilized CICD tools (Airflow, GitLab CI/CD, or Jenkins) to deploy data models to production.
Expert in Docker and Compose, including multi-stage builds and advanced networking. Manage S3 efficiently, including bucket configuration, security best practices, and cost-optimization through batch operations. Secure cloud infrastructure by implementing best practices in access control, identity and access management (IAM), and vulnerability management.
Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
Scalability and Optimization: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr ETL Developer/Snr Data Engineer/Snr DevOps Engineer/Mulesoft/AI/ML/SharePoint/K2 Developer/EDW
Responsibilities:

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.

Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM. (Java, Springboot, Hibernate, JBOSS, Microservices application and React, Redux application)

Leveraged SPARX for data modeling to support efficient pipeline development within a comprehensive enterprise architecture framework. This data model can then be used for integration with Snowflake, AWS, SAS, and Python in either a Salesforce or Mulesoft development environment.

Architected and delivered high-performance data pipelines using the Microsoft data stack (SSIS, Azure Synapse, HDInsight, ADF, Databricks) to ensure data accuracy and efficiency, empowering real-time analytics (Azure Stream Analytics) and insightful visualizations (Power BI) for improved decision-making.

Deriving business solutions through creative analysis and insights, whilst ensuring solutions are robust and fit for purpose

Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).

- Design ETL via source-to-target mapping & design documents that consider security, performance tuning & other best practices: Minimize data input, Incremental data updates, Maximize data quality, Automate, automate, automate, etc.
- Use parallel processing, Secure the data prep area, and implement DevSecOps principles for secure development practices when building ETL data pipelines.
- Deploy & test code from lower environments to production. Document technical specifications, data models, process flows etc. Perform comprehensive ETL process and other technical components to ensure data quality and functionality.
- Optimize data delivery through automation and scalable infrastructure, evaluating process improvements.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView

Designed and implemented CI/CD pipelines across Azure DevOps, GitHub Actions, and Jenkins, utilizing scripting languages (Python, Ansible, etc.) for automated tasks, to ensure continuous delivery of applications.

Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.

Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies, and Azure Security Center.

Monitored and troubleshooted Azure services using Splunk, Data Dog and Azure Network Monitor

Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Successfully migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. This involved leveraging Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration. Additionally, Azure Migrate and Azure Site Recovery ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**

Healthcare Sector: ETL Developer/Data Engineer/DevOps Engineer/ Software Test Engineer /Clouds/Linux Administrator

Responsibilities:

- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer**: Utilize Amazon EMR (Elastic MapReduce) with its Spark capabilities for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.

- **Real-time Layer**: Leverage Amazon Kinesis Data Streams and Amazon Redshift Spectrum for real-time analysis through Amazon Kinesis Data Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Utilized Matillion experience and ERwin data modeling expertise to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs.
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Using either Python or Java developed custom ETL/API orchestrations which can apply to any cloud or on-prem application.
- Implemented automated testing frameworks (Selenium, JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Collaborated with developers to ensure timely and high-quality testing.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus

Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl

Integrated AKS clusters with Azure Active Directory, Azure Key Vault, Azure Monitor, and Azure Policy for identity management, secrets management, monitoring, and governance

Optimized the performance and reliability of AKS clusters by applying best practices such as resource allocation, scaling, health checks, and logging

Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML

- Defined, developed, and documented automation technologies to improve system efficiency and streamline build, test, and deployment processes.Collaborated with development teams to implement automated build and deployment pipelines using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: ETL Developer/Data Engineer /DevOps Engineer /Software Test Engineer/SPoint Consultant /DBA
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Engineer/DevOps Engineer/Software Test Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Managed Azure resources and services, including virtual machines, storage accounts, and networking components.
- Implemented Infrastructure as Code (IaC) practices using Terraform and ARM templates for automated provisioning.
- Developed and maintained CI/CD pipelines for continuous integration and deployment of cloud-native applications.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Utilized AWS DevOps/Azure DevOps to implement robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.

- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR1, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Principle Data Engineer – SIRA – DCS – 6/5/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle Data Engineer/Senior Data Analyst /Senior DevOps Engineer/Software Engineer/Snr Linux Admin./Py /R Developer /Salesforce/Mulesoft /AI/ML/DL/BI/SharePoint/Dynamics 365 CRM Developer /EDW/DBA –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Principle Data Engineer/Snr DevOps Engineer/Python/R Developer/AI/ML/SP/D365 CRM Consultant/EDW
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Collaborate with clients & stakeholders to architect, develop, & manage databases, ETL pipelines, & management systems, ensuring end-to-end solutions align with business and user needs while meeting functional and non-functional requirements.
- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer:** Utilize Amazon EMR, a scalable managed Hadoop framework service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Manage the Commission's analytics environment, encompassing data ingestion, quality & modeling, ETL/API orchestration, visualization, metrics & reporting, and advanced research, all within a strict data governance framework.
- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Panda and Airflow or Java with Spring Batch developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Data development, optimization, and manipulation using PySpark libraries (PySpark Core, PySpark SQL, PySpark MLlib etc.) or SQL, including PySpark scripting.
- Saving a PySpark dataframe as a Delta table in a Fabric Warehouse (Lakehouse) and retrieving the same table in Metallion Warehouse (Data Lake or Delta Lake) in Auzre using spark.read.format("delta").load("<path_to_delta_table>")
- Developed ETL/ELT workflows for Azure Synapse Analytics, utilizing SSIS, SQL PolyBase on SQL Server 2022, Stream Analytics, and PowerShell scripting for optimized data movement and transformation.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Apply nested row context within calculations and conditional formatting with gradients.
- Owning the technical aspects of data and data pipelines, ensuring they adhere to data standards, architectural guidelines, and meet business requirements.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Principle Data Engineer/Snr DevOps Engineer/Python/R/Java/.Net /AI/ML/ML/SP/Power BI Developer
Responsibilities:

Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Designed and implemented a scalable Snowflake EDW (Terraform) for the hybrid Azure and AWS environment. Data from various sources (including on-premises databases) is ingested using Azure Data Factory (ADF). SSIS and Spark Notebooks handle data transformations, with Spark Notebooks leveraging Apache Spark for complex big data scenarios. This unified platform integrates with Machine Learning tools from both clouds, enabling advanced analytics and large-scale ML.

Manage the Commission's analytics environment, encompassing data ingestion, quality & modeling, ETL/API orchestration, visualization, metrics & reporting, and advanced research, all within a strict data governance framework.

Building data pipelines using Amazon EMR with PySpark to extract data from various sources (Oracle/Excel), store it in a Data Lake, and then transform and model it for analytics and reporting purposes.

Expert in data engineering and development using Databricks (Databricks Runtime, Apache Spark) on Microsoft Azure, specializing in hands-on ETL and Data Integration tasks.

Extract data from live websites into JSON format using Power BI, then utilize pandas and scikit-learn in Python for data classification and training within Jupyter Notebook. Integrate the results into Power BI dashboards or notebooks for trend prediction and visualization.

Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.

Designed and implemented IaC solutions using Terraform as the primary IaC tool. Possess experience with defining infrastructure foundations for systems or Azure products using tools like ADF, ADB, ARM templates or LAB3 Bedrock, a pre-built IaC solution leveraging Terraform for faster cloud adoption based on the MS Cloud Adoption Framework (CAF).

Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.

Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through data modeling and statistical analysis.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Utilized SPARX for data modeling to support efficient pipeline development within a comprehensive enterprise architecture framework. This data model can then be used for integration with Snowflake, AWS, SAS, and Python in either a Salesforce or Mulesoft development environment.

Deigned and implemented data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.

Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Utilized CICD tools (Airflow, GitLab CI/CD, or Jenkins) to deploy data models to production.

Expert in Docker and Compose, including multi-stage builds and advanced networking. Manage S3 efficiently, including bucket configuration, security best practices, and cost-optimization through batch operations. Secure cloud infrastructure by implementing best practices in access control, identity and access management (IAM), and vulnerability management.

Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Scalability and Optimization: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Data Engineer/Snr Data Analyst/Snr DevOps Engineer/Mulesoft/AI/ML/SharePoint/K2 Developer
Responsibilities:

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.
Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM. (Java, Springboot, Hibernate, JBOSS, Microservices application and React, Redux application)
Leveraged SPARX for data modeling to support efficient pipeline development within a comprehensive enterprise architecture framework. This data model can then be used for integration with Snowflake, AWS, SAS, and Python in either a Salesforce or Mulesoft development environment.
Architected and delivered high-performance data pipelines using the Microsoft data stack (SSIS, Azure Synapse, HDInsight, ADF, Databricks) to ensure data accuracy and efficiency, empowering real-time analytics (Azure Stream Analytics) and insightful visualizations (Power BI) for improved decision-making.
Deriving business solutions through creative analysis and insights, whilst ensuring solutions are robust and fit for purpose
Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).

- Design ETL via source-to-target mapping & design documents that consider security, performance tuning & other best practices: Minimize data input, Incremental data updates, Maximize data quality, Automate, automate, automate, etc.
- Use parallel processing, Secure the data prep area, and implement DevSecOps principles for secure development practices when building ETL data pipelines.
- Deploy & test code from lower environments to production. Document technical specifications, data models, process flows etc. Perform comprehensive ETL process and other technical components to ensure data quality and functionality.
- Optimize data delivery through automation and scalable infrastructure, evaluating process improvements.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView

Designed and implemented CI/CD pipelines across Azure DevOps, GitHub Actions, and Jenkins, utilizing scripting languages (Python, Ansible, etc.) for automated tasks, to ensure continuous delivery of applications.
Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies, and Azure Security Center.
Monitored and troubleshooted Azure services using Splunk, Data Dog and Azure Network Monitor
Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
Successfully migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. This involved leveraging Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration. Additionally, Azure Migrate and Azure Site Recovery ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Senior Data Engineer/DevOps Engineer/ Software Test Engineer /Clouds/Linux Administrator/DBA
Responsibilities:

- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer**: Utilize Amazon EMR (Elastic MapReduce) with its Spark capabilities for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams and Amazon Redshift Spectrum for real-time analysis through Amazon Kinesis Data Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Utilized Matillion experience and ERwin data modeling expertise to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs.
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Using either Python or Java developed custom ETL/API orchestrations which can apply to any cloud or on-prem application.
- Implemented automated testing frameworks (Selenium, JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Collaborated with developers to ensure timely and high-quality testing.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus

Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl

Integrated AKS clusters with Azure Active Directory, Azure Key Vault, Azure Monitor, and Azure Policy for identity management, secrets management, monitoring, and governance

Optimized the performance and reliability of AKS clusters by applying best practices such as resource allocation, scaling, health checks, and logging

Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML

- Defined, developed, and documented automation technologies to improve system efficiency and streamline build, test, and deployment processes.Collaborated with development teams to implement automated build and deployment pipelines using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Data Engineer /DevOps Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Engineer/DevOps Engineer/Software Test Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Managed Azure resources and services, including virtual machines, storage accounts, and networking components.
- Implemented Infrastructure as Code (IaC) practices using Terraform and ARM templates for automated provisioning.
- Developed and maintained CI/CD pipelines for continuous integration and deployment of cloud-native applications.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Utilized AWS DevOps/Azure DevOps to implement robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR1, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.

Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Systems & Power BI Analyst – DPHI – 5/5/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Systems & Power BI Analyst /Senior DevOps Engineer/Senior Data Engineer/Senior Clouds/Linux Administrator/ Python/R Developer /AI/ML/BI/SharePoint/D365 CRM Consultant /EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Systems & Power BI Analyst/Snr DevOps/Snr Data Engineer /AI/ML/SP/D365 CRM Consultant /EDW
Responsibilities:

- Managed and enhanced the functionality of Digital Information Office (DIO) systems and tools. Utilized Mulesoft Anypoint Studio 7 and Mule 4 for data integration, automating data exchange between disparate systems to support reporting.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including AWS API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, Dynamics 365 CRM, Power BI and SharePoint online. Additionally, support for Apache HTTP Server and IIS web servers can be integrated for SFTP service management and TLS certificate management.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, and implement security measures within the Mule flow.
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- **Integrated Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk, etc.).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
- Leverage Python for data manipulation, analysis, and integration in BI workflows, streamlining processes and managing the entire data analysis lifecycle, from data collection to model building and interpretation.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Apply nested row context within calculations and conditional formatting with gradients.
- Extract data from live websites into JSON format using Power BI, then utilize pandas and scikit-learn in Python for data classification and training within Jupyter Notebook. Integrate the results into Power BI dashboards or notebooks for trend prediction and visualization.
- Handling Power BI Deployment Pipeline (Configuration and Training to internal staff) and configuring Power BI Workspace Access (Restricted workspace for external reporting, controlled workspace for user experimentation)
- Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through extensive data modeling and statistical analysis in SQL.
- Implemented secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Programmatically interact with Power BI data and metadata using the Power BI REST API, accessible through R packages like powerbiR. Embed Power BI reports and visuals in R Shiny apps using the Power BI Embedded API and dedicated R packages like poweRbi or embedR. Utilize RStudio Server on AWS instances for remote R development and execution.
- Developed ETL/ELT workflows for Azure Synapse Analytics, utilizing SSIS, SQL PolyBase on SQL Server 2022, Stream Analytics, and PowerShell scripting for optimized data movement and transformation.

- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 environments (O365/MS Teams/SharePoint 2019/Dynamics 365 CRM /Power BI/Power Automate, etc.).
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems
- Migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL Database. Utilized Power BI Pro Get Data connectors for initial exploration and data access, followed by the Azure Database Migration Service for comprehensive database migration.
- Leveraged Azure API Management to manage both cloud-based applications via Azure Logic Apps and on-premises applications using BizTalk Server while AWS API Gateway managed MLOps deployed within the AWS environment.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Systems & Power BI Analyst/Snr DevOps/Snr Data Engineer/AI/ML/SP/Power BI Developer/EDW
Responsibilities:

Managed and enhanced the functionality of Digital Information Office (DIO) systems and tools. Utilized Mulesoft Anypoint Studio 6 and Mule 3 for data integration, automating data exchange between disparate systems to support reporting.
Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Used BizTalk Server for legacy system integration by defining data structures (schemas), mapping data formats, utilizing adapters for communication protocols (e.g., FILE, REST), & orchestrating message flows (optional) for complex scenarios.
Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2016, Dynamics 365 CRM, and SQL
Designed Amazon Redshift data warehousing solutions to facilitate data-driven decision-making.
Leverage Power Query M functional programming language to manipulate and transform data in Power Query, import, clean, transform, & combine data from various sources. Then use DAX to perform calculations & create visualizations in Power BI.
Possess advanced analytical and investigative capabilities. Demonstrated proficiency in Power BI, including the creation and sharing of reports and dashboards within the platform or via Publish to Web for broader organizational access.
Offered data-driven recommendations to meet stakeholder needs and support decision-making.
Developed interactive Power BI dashboards for visualizing key performance indicators.
Designed and developed OLAP Cubes, Dashboard & Reports and visualized data either in Power BI, Tableau or QlikView
Leveraged Azure Data Factory for data integration, enhancing insights for RBA services.
Automated data support processes through scripting in R, Python, and PowerShell 7.3.
Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
Applied statistical methods, particularly Predictive Analysis with Azure Machine Learning and AI Visuals.
Established an automated training pipeline using Google Cloud Build
Developed Business Intelligence (BI) applications and visual analytics products using Microsoft Power BI tools (like Power Query and Power Pivot) within the Azure cloud platform.
Designed bank analysis dashboards using Azure Analysis Services (AAS) as the data model and Microsoft Power BI for visualizations.
Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
Create Power BI tiles and pin them to dashboards. Develop complex DAX formulas using function calls to manipulate data.
Apply nested row context within calculations and conditional formatting with gradients.
Pin slicers or report pages to dashboards in Power BI Desktop/Service.
Extract data from live websites into JSON format using Power BI, then utilize pandas and scikit-learn in Python for data classification and training within Jupyter Notebook. Integrate the results into Power BI dashboards or notebooks for trend prediction and visualization.
Possess advanced analytical and investigative capabilities. Demonstrated proficiency in Power BI, including the creation and sharing of reports and dashboards within the platform or via Publish to Web for broader organizational access.
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage

Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Engineered ML systems using Python and machine learning frameworks (PyTorch, scikit-learn)
Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn
Showcase strong analytical and problem-solving skills in creating efficient self-learning applications.
Develop and optimize ETL processes using SQL and PySpark, ensuring seamless data integration and transformation.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Systems & Power BI Analyst/Snr DevOps/Snr Data Engineer /AI/ML/SP/K2 Developer /DBA
Responsibilities:
Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
Manage and modify systems and tools required to support DIO wide governance and reporting.
Access Tableau data programmatically from R using libraries like Tableau Server Client (TSC) for server interaction, and create interactive visualizations inspired by Tableau using packages like r2d3 or plotly.
Machine Learning with R for logistic regression user case, linear regression with user case
Design, implement & manage reporting dashboards using DIO standard tools to deliver work based on business priorities.
Manage the identification, design and implementation of internal process improvements including the automation of data synchronisation across systems.
Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
Design, develop, and maintain a robust ETL pipeline using either:
31. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
32. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
33. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Completed design, data models, loading approaches and source to target mapping documents.
Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
Created a mapping document that defines how the source data fields correspond to the target data fields.
Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline. The pipeline will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage

- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …

Employed Power BI, Tableau, or QlikView for financial analysis and visualization.
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.

- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.

Applied natural language processing (NLP) techniques such as text preprocessing, sentiment analysis, and topic modeling using the NLTK (Natural Language Toolkit) and Gensim Python libraries.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Systems Integration Developer /DevOps Engineer/Software Test Engineer/Linux Administrator/DBA
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Systems Integration Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with [Jenkins/GitLab CI/CD], developed REST APIs with Python and Java, monitored AWS cloud resources (EC2 instances and S3 buckets) using CloudWatch and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory [for secure storage and distribution of artifacts] within an agile environment, leveraging [GitLab/a similar CI/CD platform] for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Systems Integration Developer /DevOps Engineer/Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.

Database: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
Reporting: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
Clouds: MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
CMS: SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Solution Architect – Integration (Mulesoft) – NSWPF – Torch Recruitment 30/4/2024 – Interviewed on 24/5/2024 –Closed on 30/5
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Solution Architect - Integration/Snr Integration Developer/Snr DevOps Engineer/Snr Data Engineer /Java/.Net Developer /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 CRM Developer/EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Solution Architect - Integration/Snr Integration Developer/Snr DevOps/Data Engineer /Java/.Net Developer /AI/ML/SharePoint/D365 CRM Consultant /EDW
Responsibilities:

- Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, and implement security measures within the Mule flow.
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
- **Optional Post-Deployment**: Include tasks for notifications, scripts, or additional actions after deployments.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline. The pipeline will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Managed Azure cloud environments with automated infrastructure deployment (Terraform and ARM templates) and CI/CD via Azure DevOps for continuous integration and delivery of software applications, ensuring ongoing performance optimization with Azure Cost Management for cost-efficiency.
- Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.
- Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.

- Skilled Full-Stack Engineer leveraging React/TypeScript for frontend, Java/APIs for backend development and expertise in AWS cloud. Proficient in integration tools like MuleSoft and J2EE for seamless data flow using RESTful web services.
- Developing, customizing, and implementing data integration solutions for MS Dynamics 365 CRM using KingswaySoft SSIS Integration Toolkit for Microsoft Dynamics 365. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.
- Provide on-going 3rd level support and proactive monitoring of platforms and interfaces to reduce the impact of defects and related incidents with technical leadership and guidance to junior developers fostering collaborative and knowledge-sharing
- Migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. Utilized Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration.
- Managing, supporting, and troubleshooting MuleSoft, Azure IPaaS, BizTalk, SharePoint 2019, Dynamics 365 CRM, and SQL Server 2022 on Windows Server 2022, with a strong focus on system and procedure documentation.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Integration Solution Architect /Senior Integration Developer/Senior DevOps/Data Engineer/Java/.Net /AI/ML/ML/SP/Power BI Developer

Responsibilities:

Integration Solution Architect specializing in designing frameworks to integrate client's modern cloud-based software solutions with legacy on-site applications.

Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.

Used BizTalk Server for legacy system integration by defining data structures (schemas), mapping data formats, utilizing adapters for communication protocols (e.g., FILE, REST), & orchestrating message flows (optional) for complex scenarios.

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.

Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.

Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.

Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.

Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).

Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).

Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).

Optional Post-Deployment: Include tasks for notifications, scripts, or additional actions after deployments.

Leveraged Azure Blueprints to design and implement secure Virtual Private Clouds (VPCs) and integrate cloud workloads, focusing on Windows-based systems in M365 environments (O365/MS Teams/SharePoint 2016/Dynamics 365 CRM /Power BI/Power Automate, etc.). This IPaaS solution ensures seamless data exchange and application communication.

Utilized Azure DevOps for version control and continuous integration/continuous delivery (CI/CD) pipeline implementation.

Provide on-going 3rd level support and proactive monitoring of platforms and interfaces to reduce the impact of defects and related incidents with technical leadership and guidance to junior developers fostering collaborative and knowledge-sharing

Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.

Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.

Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Migrated on-premise applications to GCP cloud using Velostrata, Cloud Migrate for Compute Engine,
Integrated R with Tableau to access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn
Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
Integrated DevSecOps practices with Infrastructure as Code (IaC) security principles to automate security configurations and ensure consistent security posture across deployments.
Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

Oct. 2013 – Nov.2015 University of Canberra (UC), ACT/SYD (Working remotely from Sydney Home Office)
Gov. University: Integration Solution Architect/Sr Integration Developer/Sr DevOps/Data Engineer/Sr Systems Engineer /ML/SPoint/K2 Developer
Responsibilities:
Software Integration Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications.
Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
Led the development, implementation, and maintenance of integrations between diverse systems and applications.
Provided 3rd level technical support, consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with standards and requirements.
Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
Designed and implemented secure Virtual Private Clouds (VPCs) using Azure Blueprints for cloud workloads, enabling seamless integration with Windows-based systems in M365 environments (O365/MS Teams/SharePoint 2019/Dynamics 365 CRM /Power BI/Power Automate, etc.) (IPaaS).
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
Optimized code for ML projects using data structures, algorithms, and software architecture knowledge.
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
Applied natural language processing techniques such as text preprocessing, sentiment analysis and topic modeling using NLTK (Natural Language Toolkit) and Gensim.
Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.

Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Integration Developer /DevOps Engineer/Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Systems Integration Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.

- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Systems Integration Developer /DevOps Engineer/Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access

OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc

AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.

Database: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,

Reporting: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API

Clouds: MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,

CMS: SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.

Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.

Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security

Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve

Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Data Engineer – Amdpersant – 25/4/2024

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter

Senior Data Engineer/Senior Data Analyst /Senior DevOps Engineer/Snr Clouds/Linux Administrator /Python /R Developer /AI/ML/DL/BI/SharePoint/Dynamics 365 CRM Consultant /EDW/DBA –Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**

Gov. Health: Senior Data Engineer/Data Analyst/Snr DevOps Engineer/Python/R Developer/AI/ML/SP/D365 CRM Consultant

Responsibilities:

- Architected and Implemented hybrid cloud infrastructure architectures between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between on-prem & both cloud platforms (utilizing VPCs in AWS and VNets in Azure) using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Design, develop, and maintain a robust ETL pipeline using either:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
- Designed and implemented a scalable Snowflake EDW (Terraform) for the hybrid Azure and AWS environment. Data from various sources (including on-premises databases) is ingested using Azure Data Factory (ADF). SSIS and Spark Notebooks handle data transformations, with Spark Notebooks leveraging Apache Spark for complex big data scenarios. This unified platform integrates with Machine Learning tools from both clouds, enabling advanced analytics and large-scale ML.
- **Deployed multi-cloud databases**: Utilized Azure Cosmos DB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using Azure Cognitive Search for native retrieval methods with RAG, libraries like Faiss or Preflexity, and Azure Cosmos DB connectors for pymongo-like access. Large language models (LLMs) like GPT-4, Sora, Mistral, and Gemini 1.5 can still be integrated for further analysis.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.

- Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through extensive data modeling and statistical analysis in SQL.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Developed ETL/ELT workflows for Azure Synapse Analytics, utilizing SSIS, SQL PolyBase on SQL Server 2022, Stream Analytics, and PowerShell scripting for optimized data movement and transformation.
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Drawing meaningful conclusions and provide actionable insights; Making weekly Data Integrity Check Report for the Operations and Decision Making groups, deleting the decisions from duplicated or cancelled activities, etc.
- Handling multiple data sources and using analysis to draw meaningful conclusions and actionable insights.
- Applied statistical methods to analyze data and derive well-reasoned and defensible conclusions.
- Designed and implemented IaC solutions using Terraform as the primary IaC tool. Possess experience with defining infrastructure foundations for systems or Azure products using tools like ADF, ADB,ARM templates or LAB3 Bedrock, a pre-built IaC solution leveraging Terraform for faster cloud adoption based on the MS Cloud Adoption Framework (CAF).
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- **Perform code release using CI/CD tools with BuildKite and GitHub**: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Data Engineer/Snr Data Analyst/Snr DevOps Engineer/Py/R Developer/AI/ML/SP/BI Consultant /EDW
Responsibilities:

Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.

Provide expert advice to stakeholders, recommending and implementing ways to improve data efficiency and reliability, ensuring associated risks with deployment are documented and understood

Designed and implemented a scalable Snowflake EDW (Terraform) for the hybrid Azure and AWS environment. Data from various sources (including on-premises databases) is ingested using Azure Data Factory (ADF). SSIS and Spark Notebooks handle data transformations, with Spark Notebooks leveraging Apache Spark for complex big data scenarios. This unified platform integrates with Machine Learning tools from both clouds, enabling advanced analytics and large-scale ML.

Developed ETL/ELT workflows for Azure Synapse Analytics, utilizing SSIS, SQL PolyBase on SQL Server 2016, Stream Analytics, and PowerShell scripting for optimized data movement and transformation.

Conducted data extraction, analysis, and modeling to support data-driven decision making.

Architected and implemented a modern 3-layer lambda architecture on Azure: **Batch layer**: Utilized Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes.

Speed layer: Employed Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis.

Unified service layer: Leveraged Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics.

Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.

Deriving business solutions through creative analysis and insights, whilst ensuring solutions are robust and fit for purpose

Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Leveraged automated techniques like hyperparameter tuning, model compression, and ensemble learning to optimize models in hybrid clouds. This empowered data scientists to automatically adjust a model's settings, reduce its size, or combine predictions for better accuracy without manual intervention.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Deigned and implemented data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.

Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,

Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Utilized CICD tools (Airflow, GitLab CI/CD, or Jenkins) to deploy data models to production.

Defined and documented existing automation technologies for improved system efficiency and build/deployment.

Expert in Docker and Compose, including multi-stage builds and advanced networking. Manage S3 efficiently, including bucket configuration, security best practices, and cost-optimization through batch operations. Secure cloud infrastructure by implementing best practices in access control, identity and access management (IAM), and vulnerability management.

Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment

Automation and CI/CD: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure. Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines

Scalability and Optimization: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.

Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Data Engineer/Snr Data Analyst/Snr DevOps Engineer/Python/R/AI/ML/SharePoint/K2 Developer/DBA
Responsibilities:

Designed and implemented hybrid cloud (AWS VPCs and Azure VNETs) with Cloud VPNs for secure, encrypted tunnels data transfer. Automated infrastructure management (Terraform/Ansible) for seamless provisioning and maintenance.

Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

Built a scalable Data Lake with Terraform for ingesting data from various sources, including potentially multiple clouds. This lake stores raw data and curated datasets for easy access. It seamlessly integrates with a multi-layered Snowflake analytics platform, allowing ready-to-use ML tools from AWS & Azure to power analytics, low-latency processing, and large-scale ML.

Architected and delivered high-performance data pipelines using the Microsoft data stack (SSIS, Azure Synapse, HDInsight, ADF, Databricks) to ensure data accuracy and efficiency, empowering real-time analytics (Azure Stream Analytics) and insightful visualizations (Power BI) for improved decision-making.

Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).

Establishing relationships with stakeholders to develop trust and confidence in designing and creating data fabric that is flexible and robust

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView

Leveraged Terraform and Helm to automate deployments of Azure Kubernetes Service (AKS) clusters for microservices.

Designed and implemented CI/CD pipelines across Azure DevOps, GitHub Actions, and Jenkins, utilizing scripting languages (Python, Ansible, etc.) for automated tasks, to ensure continuous delivery of applications.

Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.

Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies, and Azure Security Center.

Monitored and troubleshooted Azure services using Splunk, Data Dog and Azure Network Monitor

Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Successfully migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. This involved leveraging Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration. Additionally, Azure Migrate and Azure Site Recovery ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.

Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0

Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)

K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Data Engineer/Data Analyst /DevOps Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Designed and implemented a scalable Snowflake enterprise data warehouse (Terraform) ingesting data from various sources (IBM i2 iBase, TM1) using ADF. SSIS/SQL transforms data. This unified platform integrates with ML tools from both AWS and Azure, enabling advanced analytics and large-scale machine learning.
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Implemented automated testing frameworks (Selenium, JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Collaborated with developers to ensure timely and high-quality testing.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Implemented automated monitoring with Prometheus and Grafana to proactively identify and resolve infrastructure and application issues across data pipelines.
- Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus

Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl

Integrated AKS clusters with Azure Active Directory, Azure Key Vault, Azure Monitor, and Azure Policy for identity management, secrets management, monitoring, and governance

Optimized the performance and reliability of AKS clusters by applying best practices such as resource allocation, scaling, health checks, and logging

Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML

- Defined, developed, and documented automation technologies to improve system efficiency and streamline build, test, and deployment processes.Collaborated with development teams to implement automated build and deployment pipelines using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Data Engineer /Data Analyst /DevOps Engineer/SharePoint Consultant /DBA
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Engineer/Data Analyst /DevOps Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Managed Azure resources and services, including virtual machines, storage accounts, and networking components.
- Implemented Infrastructure as Code (IaC) practices using Terraform and ARM templates for automated provisioning.
- Developed and maintained CI/CD pipelines for continuous integration and deployment of cloud-native applications.
- Conducted regular audits and optimizations to improve performance and cost-effectiveness of Azure environments.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Automate build, test, and deployment processes using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR1, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions

- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior DevOps Engineer – Service NSW – DCS – 18/4/2024 – Interview on 4/June/2024
What is IoC, benefit of Terraform merge, How many AWS Services you used. AWS Tower, AWS Control Tower for multiple accounts (about 50 acccountts), How many IAM roles.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Senior Data Engineer/Senior Clouds/Unix/Linux Administrator /Senior Systems Engineer /Python/R Developer /AI/ML/BI/SharePoint /D365 CRM Consultant/EDW - Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Senior DevOps Engineer/Snr Data Engineer/Python/R Developer/AI/ML/DL/BI/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architected and Implemented hybrid cloud infrastructure architectures between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between on-prem & both cloud platforms (utilizing VPCs in AWS and VNets in Azure) using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Collaboratively develop, release, maintain, and enhance product and platform services from start to finish to deliver high-quality, resilient, and reusable solutions that support customer channels.
- Designed and implemented a highly scalable, secure platform using IaC tools (CloudFormation/Terraform/AWS CDK) to support critical business services.
- **Cloud Security with AI**: Implemented AI-driven cloud security solutions utilizing McAfee ESM and Palo Alto Networks Cortex XDR, bolstering threat detection and data security while leveraging CloudWatch insights to anticipate AWS service changes for proactive security adjustments.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements and cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps

- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Leverage AI tools like OpenAI to automate repetitive tasks like code generation and testing, freeing up time to tackle the most complex challenges and architect innovative solutions for the online application.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Utilized AI/ML technologies like CrewAI and MLOps to automate tasks, improve productivity, and reduce costs in sales, customer service, and model building. Leveraged TinyML to integrate AI with IoT/mobile devices for innovative solutions.
- Leveraged AWS cloud-native services like EC2, S3, and Lambda to implement proven native design patterns for scalable data solutions, including microservices and serverless architectures.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
- Extensive expertise in programming languages such as Java, Python, Go, C#.Net & PowerShell/Shell scripting languages,
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Collaborate and automate using AWS Cost Allocation Tags for precise cost allocation and integrating cost data via the Cost Explorer API for advanced analysis. Utilize Amazon Chime for secure communication & collaboration with the FinOps team.
- Optimize resource usage: Utilize AWS Reserved Instances and Savings Plans for predictable workloads and committed use discounts. Employ AWS Compute Optimizer to right-size EC2 instances for optimal cost and performance.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or batch scheduling via AutoSys.
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Snr DevOps Engineer/Snr Data Engineer/Python/R Developer/Liunx Admin/AI/ML/SP/BI Consultant/EDW
Responsibilities:

Built and secured hybrid cloud architecture connecting AWS VPCs and GCP Cloud VPCs: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Set up and maintain the unit's platform, ensuring smooth operation of services, and resolving any issues encompassing platform engineering, service implementation, maintenance, and best practice administration for fit-for-purpose products.

Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Worked closely with Cloud Platform Security Squad to assess and adapt security mechanisms for AWS & Azure hybrid env.

Early detection: Integrate security scanning tools (e.g., SonarQube, OWASP ZAP) into the CI pipeline, reducing the time and effort required for remediation.

Secure Coding Practices: Implement secure coding practices to minimize vulnerabilities in applications.

Vulnerability Management: Regularly conduct static code analysis and dynamic application security testing to identify and address potential vulnerabilities.

Secrets Management: Utilize secrets management tools (e.g., HashiCorp Vault) to securely store and manage sensitive credentials.

Shift Left Security: By integrating security checks into CI, we follow the DevSecOps approach of "shifting left" security. This means prioritizing security throughout the development lifecycle, not just at the deployment stage.

Faster Feedback: CI pipelines typically run more frequently than CD pipelines. Integrating security scans in CI provides developers with faster feedback on potential security issues in their code.

Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.

Champion DevOps innovation, leveraging agile methodologies and tech to streamline high-quality software delivery.

Leveraged AWS cloud-native services like EC2, S3, and Lambda to implement proven native design patterns for scalable data solutions, including microservices and serverless architectures.

Efficiently perform platform engineering and application maintenance tasks using orchestration tools (Terraform, Azure ARM, AWS CloudFormation) & strategy-building tools (CloudHealth, Flexera Cloud Management) to ensure optimal performance

Understand and analyze your AWS costs by leveraging AWS Cost Explorer and Cost & Usage Reports to analyze used AWS costs, identify cost drivers, & track trends. Set budgets with AWS Budgets to monitor spending and receive alerts before exceeding limits.

Collaborate and Automate Cost Management:

34. Leverage GCP Cost Allocation Tags: Implement GCP Cost Allocation Tags for granular cost allocation. Integrate cost data with BigQuery for advanced cost analysis using the BigQuery Cost Analysis Dataset.
35. Utilize Google Meet for Secure Communication: Use Google Meet for secure communication and collaboration with the FinOps team.

Provide pre-project consulting, leveraging expertise to identify & mitigate risks, and propose solutions aligned with IT strategy & architecture. Bridge system designs to operations with high-level options, concepts, and estimates.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.

Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources

Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)

Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.

Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation and management of deployed machine learning models in production environments.

Empower the team for project success through coaching & mentoring. Provide direct support to internal/external teams across deployments and production issues.

Compile timely reports and briefings for leadership; maintain accurate records; ensure compliant and prompt responses.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer /Sr. Data Engineer/Clouds/Linux Admin./AI/ML/DL/BI/SP/K2 Developer/EDW
Responsibilities:

Designed and implemented hybrid cloud (AWS VPCs and Azure VNETs) with Cloud VPNs for secure, encrypted tunnels data transfer. Automated infrastructure management (Terraform/Ansible) for seamless provisioning and maintenance.

Achieved a comprehensive and secure development lifecycle via embedding security practices throughout the development and operations processes by implementing a robust DevSecOps approach. **Key elements** include:

Streamlined security automation: Tight integration of security checks and gates within continuous integration and continuous delivery (CI/CD) pipelines to detect and address vulnerabilities early and efficiently.

Infrastructure as Code with Terraform: Provisioning and managing infrastructure in a secure, scalable, and auditable manner using Terraform, guaranteeing consistent and auditable infrastructure configurations.

API security for data integrity and protection: Rigorous protection of application programming interfaces (APIs) through authentication, authorization, encryption, and input validation to safeguard sensitive data from unauthorized access and manipulation.

Threat modeling for proactive risk mitigation: Proactive identification and assessment of potential threats and vulnerabilities through threat modeling techniques to implement preventive measures and reduce risk exposure.

Integration and Maintenance: Integrate frameworks to avoid conflicts (consider ACSC Essential Eight as a foundation).

Ensured Cloud Security by leveraging AWS IAM for identity governance, storing container images securely in ECR, deploying them on ECS (Elastic Container Service), and utilizing Lambda for serverless execution. This establishes a secure development and container orchestration pipeline within the AWS cloud.

Worked with DevOps Engineering team to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.

Implemented **Azure Cost Management Tags** for granular cost allocation**.** Integrated **cost data** with **Azure Cost Management API for** in-depth analysis**.** Utilized secure **MS Teams** for communication with the **FinOps team.**

Implemented **Azure Advisor** for Cost Optimization**:** Leverage Azure Advisor for recommendations on optimizing resource utilization, including right-sizing Azure VMs for cost and performance balance.

Implemented and maintained AKS clusters for various microservices applications using Terraform and Helm Charts.

Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase.

Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake

Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.

Employed Amazon CloudWatch for healthy checks of AWS resources. For application log analysis, leveraged either Splunk (real-time) or the ELK Stack (large volume analysis). Additionally, Prometheus monitored system metrics and set alerts.

Unified user authentication and authorization: Implement Amazon Cognito for user identity and access management across web and mobile applications.

Docker Images: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.

Migrated on-premise applications to the cloud using AWS services, including: **AWS Direct Connect** for establishing a dedicated network connection for ongoing data transfer. **Storage Gateway** to seamlessly connect on-premises storage with AWS cloud storage. (Optional) **AWS Snowball Edge** (if used) for offline data transfer during the initial migration.

Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Quality Assurance Engineer/Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Collaborate with product engineering teams across the orgnization to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services
- Monitor upcoming AWS service changes and adapt security mechanisms accordingly.
- **Secure Secrets**: Use tools like HashiCorp Vault or AWS Secrets Manager to safeguard sensitive information.
- **Patch Regularly**: Keep systems and applications up to date with security patches.
- **Network Segmentation**: Implement network segmentation and firewalls.
- **Compliance and Scanning**: Ensure compliance with industry standards (e.g., GDPR, HIPAA) and integrate security scanning tools (e.g., SonarQube, OWASP ZAP) into your development pipeline to detect and address vulnerabilities early
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer/Assistant Quality Assurance Engineer/Software Test Engineer/SP Consultant

Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- **Preparation**: Prioritize frameworks based on the needs (PCI-DSS for card data). Conduct a gap analysis to identify areas for improvement. Allocate resources (personnel, budget) for implementation.
- **Implementation**: Develop a security policy aligned with the chosen framework. Implement missing controls (technical & administrative). Document all security policies, procedures, and risk assessments.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Sr. Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, >NET Core, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Data Engineer – Fair Trading –DCS – applied on 18/4/2024 By 19/4/24 – No response
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Data Engineer/Senior Data Analyst /Senior DevOps Engineer/Snr Clouds/Linux Administrator /Python /R Developer /AI/ML/DL/BI/SharePoint/Dynamics 365 CRM Consultant /EDW/DBA –Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Senior Data Engineer/Data Analyst/Snr DevOps Engineer/Python/R Developer/AI/ML/SP/D365 CRM Consultant
Responsibilities:

- Architected and Implemented hybrid cloud infrastructure architectures between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between on-prem & both cloud platforms (utilizing VPCs in AWS and VNets in Azure) using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Act as a trusted partner collaborating with stakeholders to design, build, test and maintain complex data management systems, to ensure they meet business requirements and user needs
- Designed and implemented a scalable Snowflake enterprise data warehouse (Terraform) ingesting data from various sources (IBM i2 iBase, TM1) using ADF. SSIS/SQL transforms data. This unified platform integrates with ML tools from both AWS and Azure, enabling advanced analytics and large-scale machine learning.
- **Deployed multi-cloud databases**: Utilized Azure Cosmos DB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using Azure Cognitive Search for native retrieval methods with RAG, libraries like Faiss or Preflexity, and Azure Cosmos DB connectors for pymongo-like access. Large language models (LLMs) like GPT-4, Sora, Mistral, and Gemini 1.5 can still be integrated for further analysis.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through extensive data modeling and statistical analysis in SQL.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Developed ETL/ELT workflows for Azure Synapse Analytics, utilizing SSIS, SQL PolyBase on SQL Server 2022, Stream Analytics, and PowerShell scripting for optimized data movement and transformation.
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Drawing meaningful conclusions and provide actionable insights; Making weekly Data Integrity Check Report for the Operations and Decision Making groups, deleting the decisions from duplicated or cancelled activities, etc.
- Handling multiple data sources and using analysis to draw meaningful conclusions and actionable insights.
- Applied statistical methods to analyze data and derive well-reasoned and defensible conclusions.
- Designed and implemented IaC solutions using Terraform as the primary IaC tool. Possess experience with defining infrastructure foundations for systems or Azure products using tools like ADF, ADB,ARM templates or LAB3 Bedrock, a pre-built IaC solution leveraging Terraform for faster cloud adoption based on the MS Cloud Adoption Framework (CAF).
- Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 4.3, Python 3.11, PSPP, JMP, JASP, EViews, StatCrunch, etc.
- Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- **Perform code release using CI/CD tools with BuildKite and GitHub**: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Data Engineer/Snr Data Analyst/Snr DevOps Engineer/Py/R Developer/AI/ML/SP/BI Consultant /EDW
Responsibilities:

Built and secured hybrid cloud architecture connecting AWS VPCs and GCP Cloud VPCs: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Provide expert advice to stakeholders, recommending and implementing ways to improve data efficiency and reliability, ensuring associated risks with deployment are documented and understood

Designed and implemented a scalable Snowflake enterprise data warehouse (Terraform) ingesting data from various sources (IBM i2 iBase, TM1) using ADF. SSIS/SQL transforms data. This unified platform integrates with ML tools from both AWS and Azure, enabling advanced analytics and large-scale machine learning.

Perform data extraction and analysis including data modeling and statistical analysis to assist with data-driven decision making for current and future projects.

Developed ETL/ELT workflows for Azure Synapse Analytics, utilizing SSIS, SQL PolyBase on SQL Server 2016, Stream Analytics, and PowerShell scripting for optimized data movement and transformation.

Conducted data extraction, analysis, and modeling to support data-driven decision making.

Architected and implemented a modern 3-layer lambda architecture on Azure: **Batch layer**: Utilized Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes.

Speed layer: Employed Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis.

Unified service layer: Leveraged Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics.

Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.

Deriving business solutions through creative analysis and insights, whilst ensuring solutions are robust and fit for purpose

Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Leveraged automated techniques like hyperparameter tuning, model compression, and ensemble learning to optimize models in hybrid clouds. This empowered data scientists to automatically adjust a model's settings, reduce its size, or combine predictions for better accuracy without manual intervention.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Deigned and implemented data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.

Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,

Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.

Utilized CICD tools (Airflow, GitLab CI/CD, or Jenkins) to deploy data models to production.

Defined and documented existing automation technologies for improved system efficiency and build/deployment.

Expert in Docker and Compose, including multi-stage builds and advanced networking. Manage S3 efficiently, including bucket configuration, security best practices, and cost-optimization through batch operations. Secure cloud infrastructure by implementing best practices in access control, identity and access management (IAM), and vulnerability management.

Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment

Automation and CI/CD: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure. Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines

Scalability and Optimization: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Data Engineer/Snr Data Analyst/Snr DevOps Engineer/Python/R/AI/ML/SharePoint/K2 Developer/DBA
Responsibilities:

Designed and implemented hybrid cloud (AWS VPCs and Azure VNETs) with Cloud VPNs for secure, encrypted tunnels data transfer. Automated infrastructure management (Terraform/Ansible) for seamless provisioning and maintenance.

Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

Built a scalable Data Lake with Terraform for ingesting data from various sources, including potentially multiple clouds. This lake stores raw data and curated datasets for easy access. It seamlessly integrates with a multi-layered Snowflake analytics platform, allowing ready-to-use ML tools from AWS & Azure to power analytics, low-latency processing, and large-scale ML.
Responding to client requests for data products and advice while balancing competing demands to ensure objectives are achieved
Architected and delivered high-performance data pipelines using the Microsoft data stack (SSIS, Azure Synapse, HDInsight, ADF, Databricks) to ensure data accuracy and efficiency, empowering real-time analytics (Azure Stream Analytics) and insightful visualizations (Power BI) for improved decision-making.
Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).
Establishing relationships with stakeholders to develop trust and confidence in designing and creating
data fabric that is flexible and robust

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView

Leveraged Terraform and Helm to automate deployments of Azure Kubernetes Service (AKS) clusters for microservices.
Designed and implemented CI/CD pipelines across Azure DevOps, GitHub Actions, and Jenkins, utilizing scripting languages (Python, Ansible, etc.) for automated tasks, to ensure continuous delivery of applications.
Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies, and Azure Security Center.
Monitored and troubleshooted Azure services using Splunk, Data Dog and Azure Network Monitor
Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
Successfully migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. This involved leveraging Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration. Additionally, Azure Migrate and Azure Site Recovery ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0
Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)
K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Data Engineer/Data Analyst /DevOps Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Implemented automated testing frameworks (Selenium, JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Collaborated with developers to ensure timely and high-quality testing.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Implemented automated monitoring with Prometheus and Grafana to proactively identify and resolve infrastructure and application issues across data pipelines.
- Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus

Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl

Integrated AKS clusters with Azure Active Directory, Azure Key Vault, Azure Monitor, and Azure Policy for identity management, secrets management, monitoring, and governance

Optimized the performance and reliability of AKS clusters by applying best practices such as resource allocation, scaling, health checks, and logging

Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML

- Defined, developed, and documented automation technologies to improve system efficiency and streamline build, test, and deployment processes.Collaborated with development teams to implement automated build and deployment pipelines using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Data Engineer /Data Analyst /DevOps Engineer/SharePoint Consultant /DBA
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Engineer/Data Analyst /DevOps Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Managed Azure resources and services, including virtual machines, storage accounts, and networking components.
- Implemented Infrastructure as Code (IaC) practices using Terraform and ARM templates for automated provisioning.
- Developed and maintained CI/CD pipelines for continuous integration and deployment of cloud-native applications.
- Conducted regular audits and optimizations to improve performance and cost-effectiveness of Azure environments.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Automate build, test, and deployment processes using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.

- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR1, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Systems Administrator – DPHI – 19/4/24
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Systems Administrator/Senior Clouds/Linux Administrator/Senior DevOps Engineer/Senior Data Engineer /AI/ML/BI/SharePoint/D365 CRM Consultant /EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Systems Administrator/Senior DevOps/Senior Data Engineer /AI/ML/SP/D365 CRM Consultant /EDW
Responsibilities:

- Systems Administration and Configuration of cross-platform technology solutions for hybrid (AWS/Azure) & on-premise applications, including planning, development, installation, configuration, maintenance, & support of complex ICT systems.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including AWS API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Java Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM. Additionally, support for Apache HTTP Server and IIS web servers can be integrated for SFTP service management and TLS certificate management.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, and implement security measures within the Mule flow.
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- **Integrated Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk, etc.).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
- Implemented secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Engineer and productionise ML models for real-world impact, leveraging the Python, AWS stack and top-notch MLOps tools.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 environments (O365/MS Teams/SharePoint 2019/Dynamics 365 CRM /Power BI/Power Automate, etc.).
- Proficiently managed key services include Active Directory, both cloud and on premise, modern authentication technologies, Azure MFA/single sign on, InTune, certificate services, remote access and identity services including B2B and B2C tenants
- Managed Windows server 2019/2022 services, RADIUS, Remote access, trusts, group policies, DHCP, Enterprise CA, DNS and network device enrolment services using InTune/End of point management /MCCM, Azure Monitor, Ngaios, etc.
- Proven ability to manage, provide 3rd level support, & troubleshoot on-premises solutions like SharePoint 2019, Dynamics 365 CRM, & SQL Server 2022 on Windows Server 2022, along with extensive experience designing, implementing, and configuring Azure & Office 365 services (including Azure Files, D365 CRM, Power BI, Power Apps, Power Automate, SPoint & Exchange Online). This demonstrates a strong focus on system and procedure documentation for both environments.
- Provided day-to-day ICT technical support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Using InTune and Configuration Service Providers(CSP) for mobile device management (MDM) on Windows devices.
- Migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL Database. Utilized Power BI Pro Get Data connectors for initial exploration and data access, followed by the Azure Database Migration Service for comprehensive database migration.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems
- Managed IT infrastructure & EDW data backup using Commvault Data Protection /Veeam Backup for Microsoft 365 v6.0/7.0
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Leveraged Azure API Management to manage both cloud-based applications via Azure Logic Apps and on-premises applications (potentially using BizTalk Server or other integration tools) while AWS API Gateway managed MLOps deployed within the AWS environment.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.
- Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Senior Systems Administrator/Senior Clouds/DevOps Engineer /ML/SP/Power BI Developer/EDW
Responsibilities:

Administer, support, maintain and monitor Windows Server fleet (VMWare/AWS/Azure) systems, and in addition provide web server system administration and technical support as required for the Digital Information Office (DIO) Business.

Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Provides technical support and systems administration for the day-to-day operations of the organisation's ICT systems.
Used BizTalk Server for legacy system integration by defining data structures (schemas), mapping data formats, utilizing adapters for communication protocols (e.g., FILE, REST), & orchestrating message flows (optional) for complex scenarios.
Providing enterprise level systems administration of identity and directory services within a Microsoft 365 environment.
Proficiently managed Microsoft Windows 2016 server services, including Active Directory, both cloud and on-premise, RADIUS, trusts, group policies, remote access, DHCP, Enterprise CA, DNS, and network device enrolment services.
Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2016, Dynamics 365 CRM, and SQL Server 2016 on Windows Server 2016, with a strong focus on system and procedure documentation.
Provided essential day-to-day ICT support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
Implemented and Administered modern authentication technologies and InTune for efficient mobile management
Administer, support, troubleshoot SharePoint 2016, D365 CRM, SQL Server 2016 on Windows server 2016 OS platform.
Using InTune and Configuration Service Providers(CSP) for mobile device management (MDM) on Windows devices.
Managed multi-domain environments in a federated setting, focusing on modern authentication, file sharing services, etc.
Administered certificate services and remote access solutions. Provided technical support for B2B and B2C tenants
Managing, supporting, troubleshooting SharePoint 2016, D365 CRM, SQL Server 2016 on Windows server 2022 OS platform. Experiencing in creation and maintenance of system and procedure documentation
Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Engineered ML systems using Python and machine learning frameworks (PyTorch, scikit-learn)
Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn
Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git
Showcase strong analytical and problem-solving skills in creating efficient self-learning applications.
Develop and optimize ETL processes using SQL and PySpark, ensuring seamless data integration and transformation.
Deploy machine learning solutions for various clients using IBM's Watson Studio platform.
Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Systems Administrator/Senior Clouds/DevOps Engineer /ML/SharePoint /K2 Developer /DBA
Responsibilities:

Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
Administer and maintain Active Directory, both cloud and on-premise, with a strong focus on modern authentication technologies, Azure MFA, and single sign-on. Manage InTune/SCCM, identity services, including B2B and B2C tenants.
Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
Proficiently managed Microsoft Windows 2016 server services, including Active Directory, both cloud and on-premise, RADIUS, trusts, group policies, remote access, DHCP, Enterprise CA, DNS, and network device enrolment services.
Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2016, Dynamics 365 CRM, and SQL Server 2016 on Windows Server 2016, with a strong focus on system and procedure documentation.
Provided essential day-to-day ICT support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.

Implemented and Administered modern authentication technologies and InTune for efficient mobile management
Using InTune and Configuration Service Providers(CSP) for mobile device management (MDM) on Windows devices.
Manage, support, troubleshoot SharePoint 2013, Dynamics CRM 2013, SQL Server 2012R2 on Windows server 2012R2 OS platform.
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage

- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …

Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline. The pipeline will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Optimized code for ML projects using data structures, algorithms, and software architecture knowledge.
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
Applied natural language processing (NLP) techniques such as text preprocessing, sentiment analysis, and topic modeling using the NLTK (Natural Language Toolkit) and Gensim Python libraries.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior Systems Administrator/Senior Clouds/DevOps Engineer /SharePoint Consultant /DBA
Responsibilities:

- Administer, support, maintain and monitor Windows Server fleet (VMWare/AWS/Azure) systems, and in addition provide web server system administration and technical support as required for the Digital Information Office (DIO) Business.
- Proficiently managed Microsoft Windows 2012 server services, including Active Directory, both cloud and on-premise, RADIUS, trusts, group policies, remote access, DHCP, Enterprise CA, DNS, and network device enrolment services.
- Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2013, Dynamics 365 CRM, and SQL Server 2013 on Windows Server 2013, with a strong focus on system and procedure documentation.
- Provided essential day-to-day ICT support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Implemented and Administered modern authentication technologies and InTune for efficient mobile management
- Successfully managed and maintained the Standard Operating Environment (SOE) for over 400 workstations, ensuring system stability and optimal performance.
- Windows Server 2012 OS administration, providing support for services like DNS and DHCP, and contributing to network service maintenance. Contributed to the maintenance of enterprise CA and other network services
- Manage, support, troubleshoot SharePoint 2013, Dynamics CRM 2013, SQL Server 2012R2 on Windows server 2012R2 OS platform. Collaborated with IT Security to ensure SOE security hardening in compliance with industry standards.
- Offered 3rd level technical support through various channels, including phone, email, and in-person interactions, resolving hardware and software issues, while also installing and configuring software and hardware.

- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Systems Administrator (AD/SCCMSCOM)/Systems Engineer/Data Engineer/SharePoint Consultant
Responsibilities:

- Pprovides technical support and system administration for the day-to-day operations of the organisation's ICT systems.
- Proficiently managed Microsoft Windows 2012 server services, including Active Directory, both cloud and on-premise, RADIUS, trusts, group policies, remote access, DHCP, Enterprise CA, DNS, and network device enrolment services.
- Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2013, Dynamics 365 CRM, and SQL Server 2012 on Windows Server 2012, with a strong focus on system and procedure documentation.
- Provided essential day-to-day ICT support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Implemented and Administered modern authentication technologies and InTune for efficient mobile management
- Provided 3rd level desktop, systems, and network support, diagnosing and resolving complex technical issues.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR) option, etc.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /ICT System Administrator (AD/SCCM/SCOM)/DBA
Responsibilities:

- Conducted the installation, configuration, and troubleshooting of hardware, software, and networking equipment.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate "zone1", "WWN1","WWN2"; Create a zoneset configuration by #cfgcreate "test_cfg", "zone1", "zone2"; Add a zone to a zone configuration via cfgadd, etc.
- Installed and configured a WebLogic domain with a frontend Apache Web Server acting as a reverse proxy. The backend integrates Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008 /BDD2007 Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /ICT System Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.

- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Installed and configured a WebLogic domain with a frontend Apache Web Server acting as a reverse proxy. The backend integrates Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /ICT Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patching for Operating Systems and Applications
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Proficiently handled hardware, software, and networking equipment installation and configuration. Provided effective support for Office Online Specialist functions, swiftly resolving technical issues while managing Active Directory and assisting in AD migrations between forests.
- Conducted installation, configuration, and testing of hardware and software solutions as well as routine maintenance tasks
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers & F5 load balancers v9
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java/Kotlin, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; MuleSoft Anypoint Studio and Mule 4, BizTalk Server 1.0~2020, RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
Reporting: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
Clouds: MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
CMS: SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Integration Developer – Department of Planning, Housing and Infrastructure DPHI – 7/4/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Integration Developer/Senior DevOps Engineer/Snr Data Engineer/Java/.Net Developer/Clouds/Linux Administrator /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 CRM Devloper/EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Senior Integration Developer/Sr DevOps/Data Engineer/Java/.Net Developer/AI/ML/SP/D365 CRM Consultant
Responsibilities:

- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, and implement security measures within the Mule flow.
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
- **Optional Post-Deployment**: Include tasks for notifications, scripts, or additional actions after deployments.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline. The pipeline will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Managed Azure cloud environments with automated infrastructure deployment (Terraform and ARM templates) and CI/CD via Azure DevOps for continuous integration and delivery of software applications, ensuring ongoing performance optimization with Azure Cost Management for cost-efficiency.
- Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.
- Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
- Skilled Full-Stack Engineer leveraging React/TypeScript for frontend, Java/APIs for backend development and expertise in AWS cloud. Proficient in integration tools like MuleSoft and J2EE for seamless data flow using RESTful web services.
- Developing, customizing, and implementing data integration solutions for MS Dynamics 365 CRM using KingswaySoft SSIS Integration Toolkit for Microsoft Dynamics 365. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.
- Provide on-going 3rd level support and proactive monitoring of platforms and interfaces to reduce the impact of defects and related incidents with technical leadership and guidance to junior developers fostering collaborative and knowledge-sharing
- Migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. Utilized Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration.
- Managing, supporting, and troubleshooting MuleSoft, Azure IPaaS, BizTalk, SharePoint 2019, Dynamics 365 CRM, and SQL Server 2022 on Windows Server 2022, with a strong focus on system and procedure documentation.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Senior Integration Developer/Senior DevOps/Data Engineer/Java/.Net /AI/ML/ML/SP/Power BI Developer
Responsibilities:

- Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Used BizTalk Server for legacy system integration by defining data structures (schemas), mapping data formats, utilizing adapters for communication protocols (e.g., FILE, REST), & orchestrating message flows (optional) for complex scenarios.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, and implement security measures within the Mule flow.
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
- **Optional Post-Deployment**: Include tasks for notifications, scripts, or additional actions after deployments.
- Leveraged Azure Blueprints to design and implement secure Virtual Private Clouds (VPCs) and integrate cloud workloads, focusing on Windows-based systems in M365 environments (O365/MS Teams/SharePoint 2016/Dynamics 365 CRM /Power BI/Power Automate, etc.). This IPaaS solution ensures seamless data exchange and application communication.
- Utilized Azure DevOps for version control and continuous integration/continuous delivery (CI/CD) pipeline implementation.
- Provide on-going 3rd level support and proactive monitoring of platforms and interfaces to reduce the impact of defects and related incidents with technical leadership and guidance to junior developers fostering collaborative and knowledge-sharing
- Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
- Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Migrated on-premise applications to GCP cloud using Velostrata, Cloud Migrate for Compute Engine,
- Integrated R with Tableau to access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
- Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- **Integrated DevSecOps practices with Infrastructure as Code (IaC) security principles** to automate security configurations and ensure consistent security posture across deployments.
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Sr Integration Developer/Sr DevOps/Data Engineer/Sr Systems Engineer/ML/SPoint/K2 Developer
Responsibilities:

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
Led the development, implementation, and maintenance of integrations between diverse systems and applications.
Provided 3rd level technical support, consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with standards and requirements.
Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
Designed and implemented secure Virtual Private Clouds (VPCs) using Azure Blueprints for cloud workloads, enabling seamless integration with Windows-based systems in M365 environments (O365/MS Teams/SharePoint 2019/Dynamics 365 CRM /Power BI/Power Automate, etc.) (IPaaS).
Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
Optimized code for ML projects using data structures, algorithms, and software architecture knowledge.
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
Applied natural language processing techniques such as text preprocessing, sentiment analysis and topic modeling using NLTK (Natural Language Toolkit) and Gensim.
Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Integration Developer /DevOps Engineer/Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Systems Integration Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Systems Integration Developer /DevOps Engineer/Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Responsible for integration development process, ensuring adherence to standards and documentation of requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
Reporting: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
Clouds: MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
CMS: SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve

Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Azure DevOps Engineer – Rec4Tech – 23/3/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Software Engineer/Sr. Data Engineer/Analyst /Clouds/Linux Administrator /Python /R Developer /AI/ML/DL/BI/SharePoint/Dynamics 365 CRM Consultant /EDW/DBA –Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Principle DevOps Engineer/Software Engineer/Sr. Data Engineer/Py/R Dev/AI/ML/SP/D365 CRM Consultant
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Play a pivotal role within the software team, focusing on delivering robust, reliable, and secure cloud capabilities.
- Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.
- **Architected and implemented a modern 3-layer lambda architecture on Azure**: **Batch layer**: Utilized Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes.
- **Speed layer**: Employed Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and implemented IaC solutions using Terraform as the primary IaC tool. Possess experience with defining infrastructure foundations for systems or Azure products using tools like ADF, ADB,ARM templates or LAB3 Bedrock, a pre-built IaC solution leveraging Terraform for faster cloud adoption based on the MS Cloud Adoption Framework (CAF).
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Built secure and standardized cloud infrastructure with Terraform and Packer for VM deployments. Implemented automated monitoring and alerting for proactive issue identification, using Ansible for post-provisioning configuration & security scans.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements and cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- **Perform code release using CI/CD tools with BuildKite and GitHub**: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …

- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Bridge the gap between R and low-level C++ for precise control and data manipulation.
- Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R with RStudio and Rccp package.
- Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Sr. DevOps Engineer/Software Engineer/Sr. Data Engineer/Py/R Developer/AI/ML/SP/Consultant /EDW
Responsibilities:

Built and secured **hybrid cloud architecture between AWS and GCP**: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/ J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Designed, built, & enhanced high-performance computing for processing & analyzing market data pipelines on RHEL7 in AWS. Experimented with novel algorithms and techniques, resulting in significant model performance improvements.

Leveraged automated techniques like hyperparameter tuning, model compression, and ensemble learning to optimize models in hybrid clouds. This empowered data scientists to automatically adjust a model's settings, reduce its size, or combine predictions for better accuracy without manual intervention.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Deigned and implemented data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.

Collaborate with data scientists and engineers on the ML lifecycle: problem definition, data exploration, model selection (classification, regression, clustering), and implementation for scalability.

Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,

Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.

Utilized CICD tools (Airflow, GitLab CI/CD, or Jenkins) to deploy data models to production.

Defined and documented existing automation technologies for improved system efficiency and build/deployment.

Maintained clear visibility of version control between production systems and planned deployments for enhanced release management using Plastic SCM for precise version control. Achieved consistency and reproducibility across environments.

Defined, developed, and documented automation technologies to improve system efficiency and streamline build, test, and deployment processes.

Managed Linux servers using configuration management tools like Ansible, Chef, and Puppet. Used Maven, ANT, and Gradle for building Java projects.

Expert in Docker and Compose, including multi-stage builds and advanced networking. Manage S3 efficiently, including bucket configuration, security best practices, and cost-optimization through batch operations. Secure cloud infrastructure by implementing best practices in access control, identity and access management (IAM), and vulnerability management.

Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment

Automation and CI/CD: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure. Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines

Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.

Scalability and Optimization: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Azure DevOps Engineer/Software Engineer/Sr. Data Engineer/Python/R/AI/ML/SP/K2 Developer
Responsibilities:

Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.

Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

Built a scalable Data Lake with Terraform for ingesting data from various sources, including potentially multiple clouds. This lake stores raw data and curated datasets for easy access. It seamlessly integrates with a multi-layered Snowflake analytics platform, allowing ready-to-use ML tools from AWS & Azure to power analytics, low-latency processing, and large-scale ML.

Lead transformative initiatives, contributing to the evolution of their existing products and the development of new ones.

Collaborate with data scientists and engineers to design and implement scalable ML solutions from define the problem and data, frame the ML problem: Is it classification (e.g., spam detection), regression (e.g., sales prediction), or clustering (e.g., customer segmentation)? Identify data sources to explore and clean data.

Managed Azure cloud environments with automated infrastructure deployment (Terraform and ARM templates) and CI/CD via Azure DevOps for continuous integration and delivery of software applications, ensuring ongoing performance optimization with Azure Cost Management for cost-efficiency.

Leveraged Terraform and Helm to automate deployments of Azure Kubernetes Service (AKS) clusters for microservices.

Designed and implemented CI/CD pipelines across Azure DevOps, GitHub Actions, and Jenkins, utilizing scripting languages (Python, Ansible, etc.) for automated tasks, to ensure continuous delivery of applications.

Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.

Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies, and Azure Security Center.

Monitored and troubleshooted Azure services using Splunk, Data Dog and Azure Network Monitor

Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Successfully migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. This involved leveraging Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration. Additionally, Azure Migrate and Azure Site Recovery ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.

Developed applications in Java, Microsoft .NET, and Oracle-deployed applications, ensuring seamless integration with the DevOps workflow.

Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0

Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)

K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Leveraged DataRobot's automated ML platform and best practices (Agile, DevOps, Configuration-As-Code) to develop and deploy end-to-end machine learning solutions for clients.

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**

Healthcare Sector: DevOps Engineer/Software Test Engineer/Cloud/Linux Administrator/DBA

Responsibilities:

- Led platform decisions for a greenfield project on SaaS platform on Azure. Designed & implemented CI/CD pipelines (Azure DevOps) for accelerated development & deployments. Automated SDLC with monitoring for high-quality code delivery.
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Implemented automated testing frameworks (Selenium, JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Collaborated with developers to ensure timely and high-quality testing.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Implemented automated monitoring with Prometheus and Grafana to proactively identify and resolve infrastructure and application issues across data pipelines.
- Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus

Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl

Integrated AKS clusters with Azure Active Directory, Azure Key Vault, Azure Monitor, and Azure Policy for identity management, secrets management, monitoring, and governance

Optimized the performance and reliability of AKS clusters by applying best practices such as resource allocation, scaling, health checks, and logging

Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML

Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.

- Defined, developed, and documented automation technologies to improve system efficiency and streamline build, test, and deployment processes.Collaborated with development teams to implement automated build and deployment pipelines using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia

Government Council: DevOps Engineer/Software Test Engineer/SharePoint Consultant /DBA

Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Actively contribute to and guide innovation in DevOps practices and incorporate agile delivery approaches and technologies that assist with the development and delivery of high-quality software
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 Manulife Financial Corporation (MLJ), Tokyo

Finance/Insurance Domain: Azure DevOps Engineer/Data Engineer/Senior Systems Engineer /DBA

Responsibilities:

- Managed Azure resources and services, including virtual machines, storage accounts, and networking components.
- Implemented Infrastructure as Code (IaC) practices using Terraform and ARM templates for automated provisioning.

- Developed and maintained CI/CD pipelines for continuous integration and deployment of cloud-native applications.
- Conducted regular audits and optimizations to improve performance and cost-effectiveness of Azure environments.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Automate build, test, and deployment processes using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR1, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP

CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer – EKS, AKS GKS – Akkodis – 17/3/2024 - Missed Telstra as they received 160 applications and lots of them didn't open by Ms Isha Khamma 047 7745 119 ACT Mr Teg called me on 2/5/24 for DM- role. He will pass my CV to Isha ' managr for next time similar roles
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Senior Data Engineer/Senior Clouds/Unix/Linux Administrator /Senior Systems Engineer /Python/R Developer /AI/ML/BI/SharePoint /D365 CRM Consultant/EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Principle DevOps Engineer/Sr. Data Engineer/Python/R Developer/AI/ML/DL/BI/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Leverage an SCM like Git to manage code versions and integrate a CI/CD pipeline (e.g., AWS CodePipeline, Azure DevOps, GCP Cloud Build) on your chosen cloud platform to automate deployments and streamline Devops workflows.+
- Test controls, conduct security awareness training, and continuously monitor. Utilize a Plan-Do-Check-Act (PDCA) cycle for ongoing improvement.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements and cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Leverage Infrastructure and Configuration as Code (IaC/CaC) tools like Ansible or Terraform to automate infrastructure provisioning and software configuration on your chosen cloud platform (AWS, Azure, GCP), enabling repeatable deployments and minimizing manual configuration drift.
- Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.

- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
- Streamlined the DevOps workflow on AWS using Atlassian tools: Jira, Confluence, Bitbucket, Bamboo, Bash/Shell, etc.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or batch scheduling via AutoSys.
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer/Sr. Data Engineer/Python/R Developer/Liunx Admin/AI/ML/BI/SP Consultant
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Worked with DevOps Engineering team in Revenue Digital Branch to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
Frequently implemented AWS cloud-native design patterns. Such as highly scalable and fault-tolerant **Amazon DynamoDB Amazon SQS** to send messages between microservices, or to queue tasks for later processing. **SNS** to send messages to a large number of subscribers or to send notifications to customers, or to trigger workflows in other applications.
Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.
Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.
DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
Delivered and managed multi-cloud networks, enhancing operational efficiency and network reliability.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation and management of deployed machine learning models in production environments.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Implemented and maintained AKS clusters for various microservices applications using Terraform and Helm Charts.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Either using BMC Control-M v8.0 to manage, multiple data transformation jobs or batch scheduling via AutoSys
Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer /Sr. Data Engineer/Clouds/Linux Admin. /AI/ML/DL/BI/SP/K2 Developer/EDW
Responsibilities:
Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
Worked with DevOps Engineering team to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Implemented AWS cloud-native design patterns, including leveraging Amazon CloudWatch for comprehensive monitoring and observability of AWS resources. This includes tracking metrics, logs, and events from both applications & infrastructure. Additionally, utilized Amazon Cognito for robust user identity and access management in web and mobile applications.
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.
Leverage AWS services for cloud-native design patterns:
Highly scalable and fault-tolerant monitoring: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
Unified user authentication and authorization: Implement Amazon Cognito for user identity and access management across web and mobile applications.
Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
Docker Images: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
Completed design documentation, data models, loading approaches and source to target mapping documents.
Leveraged technical expertise to consult on projects throughout their lifecycle. Conducted pre-project assessments to identify and mitigate risks, interdependencies, and exceptions. Developed solutions aligned with ICT strategy and architecture. Authored technical documentation to support new and existing applications.
Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevSecOps best practices to IT and business stakeholders.
Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.
Migrated on-premise applications to the cloud using AWS Direct Connect, Storage Gateway and AWS Snowball Edge.

Streamlined the DevOps workflow on AWS using Atlassian tools: Jira, Confluence, Bitbucket, Bamboo, Bash/Shell, etc.
Executed implementation, documentation, integration, and support operations of a variety of REST API projects
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Quality Assurance Engineer/Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve software features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer/Assistant Quality Assurance Engineer/Software Test Engineer/SP Consultant
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Actively contribute to and guide innovation in DevOps practices and incorporate agile delivery approaches and technologies that assist with the development and delivery of high-quality software
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Sr. Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access

OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc

AI, ML, DL, BI IDE: Secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.), Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, MATLAB, etc.

Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL

Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP

CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.

Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.

Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5

Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve

Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

AWS DevOps Engineer – Quay Apt. 17/3/24 Macq. Park, need change train 3 times.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Senior Data Engineer/Senior Clouds/Unix/Linux Administrator /Senior Systems Engineer /Python/R Developer /AI/ML/BI/SharePoint /D365 CRM Consultant/EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Principle DevOps Engineer/Sr. Data Engineer/Python/R Developer/AI/ML/DL/BI/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Manage and optimize cloud workloads through automation scripts, maintaining infrastructure with operations, security fixes, and keeping costs in check – handling everything from creation to ongoing care.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- **Preparation**: Prioritize frameworks based on the needs (PCI-DSS for card data). Conduct a gap analysis to identify areas for improvement. Allocate resources (personnel, budget) for implementation.
- **Implementation**: Develop a security policy aligned with the chosen framework. Implement missing controls (technical & administrative). Document all security policies, procedures, and risk assessments.
- **Integration and Maintenance**: Integrate frameworks to avoid conflicts (consider ACSC Essential Eight as a foundation).
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements and cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.

- Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.
- Using Cloud-native tools Terraform for infrastructure provisioning, Ansible for configuration management, Packer for image building, Azure DevOps for CI/CD pipelines, Splunk for log management, and Dynatrace for application monitoring to create a cohesive DevOps workflow for infrastructure and application deployment and management.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or batch scheduling via AutoSys.
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. AWS DevOps Engineer/Sr. Data Engineer/Python/R Developer/Liunx Admin/AI/ML/BI/SP Consultant
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Worked with DevOps Engineering team in Revenue Digital Branch to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.

Frequently implemented AWS cloud-native design patterns. Such as highly scalable and fault-tolerant **Amazon DynamoDB Amazon SQS** to send messages between microservices, or to queue tasks for later processing. **SNS** to send messages to a large number of subscribers or to send notifications to customers, or to trigger workflows in other applications.

Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.

Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.

Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake

Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources

Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.

Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.

Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources

Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)

Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.

Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.

DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.

Delivered and managed multi-cloud networks, enhancing operational efficiency and network reliability.

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.

Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation and management of deployed machine learning models in production environments.

Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center

Implemented and maintained AKS clusters for various microservices applications using Terraform and Helm Charts.

Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase.

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau

Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Either using BMC Control-M v8.0 to manage, multiple data transformation jobs or batch scheduling via AutoSys

Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)

Gov. University: Sr. AWS DevOps Engineer /Sr. Data Engineer/Clouds/Linux Admin. /AI/ML/DL/BI/SP/K2 Developer/EDW

Responsibilities:

Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.

Work closely with development teams to ensure proper integration of applications with cloud services.

Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Implemented AWS cloud-native design patterns, including leveraging Amazon CloudWatch for comprehensive monitoring and observability of AWS resources. This includes tracking metrics, logs, and events from both applications & infrastructure. Additionally, utilized Amazon Cognito for robust user identity and access management in web and mobile applications.

Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.

Leverage AWS services for cloud-native design patterns:

Using Cloud-native tools Terraform for infrastructure provisioning, Ansible for configuration management, Packer for image building, Azure DevOps for CI/CD pipelines, Splunk for log management, and Dynatrace for application monitoring to create a cohesive DevOps workflow for infrastructure and application deployment and management.

Unified user authentication and authorization: Implement Amazon Cognito for user identity and access management across web and mobile applications.

Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.

Zhi Ming ZHANG - Australian Citizen File Ref: ZMZITILJP003
1303 / 29 Hunter St.| Parramatta 2150 | NSW Mobile: 0402-997-679 | mingjp@gmail.com

Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.

Docker Images: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Completed design documentation, data models, loading approaches and source to target mapping documents.

Leveraged technical expertise to consult on projects throughout their lifecycle. Conducted pre-project assessments to identify and mitigate risks, interdependencies, and exceptions. Developed solutions aligned with ICT strategy and architecture. Authored technical documentation to support new and existing applications.

Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevSecOps best practices to IT and business stakeholders.

Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.

Migrated on-premise applications to the cloud using AWS Direct Connect, Storage Gateway and AWS Snowball Edge.

Streamlined the DevOps workflow on AWS using Atlassian tools: Jira, Confluence, Bitbucket, Bamboo, PowerShell & Bash.

Executed implementation, documentation, integration, and support operations of a variety of REST API projects

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Quality Assurance Engineer/Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve software features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer/Assistant Quality Assurance Engineer/Software Test Engineer/SP Consultant
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

- Actively contribute to and guide innovation in DevOps practices and incorporate agile delivery approaches and technologies that assist with the development and delivery of high-quality software
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Sr. Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Principle DevOps Engineer – DCS Syd 17/3/24
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Senior Data Engineer/Senior Clouds/Unix/Linux Administrator /Senior Systems Engineer /Python/R Developer /AI/ML/BI/SharePoint /D365 CRM Consultant/EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Principle DevOps Engineer/Sr. Data Engineer/Clouds/Linux Admin. /AI/ML/SPoint/D365 CRM Consultant /EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- **Preparation**: Prioritize frameworks based on the needs (PCI-DSS for card data). Conduct a gap analysis to identify areas for improvement. Allocate resources (personnel, budget) for implementation.
- **Implementation**: Develop a security policy aligned with the chosen framework. Implement missing controls (technical & administrative). Document all security policies, procedures, and risk assessments.
- **Integration and Maintenance**: Integrate frameworks to avoid conflicts (consider ACSC Essential Eight as a foundation).

- Test controls, conduct security awareness training, and continuously monitor. Utilize a Plan-Do-Check-Act (PDCA) cycle for ongoing improvement.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements and cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or batch scheduling via AutoSys.
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Principle DevOps Engineer/Sr. Data Engineer/Python/R Developer/Liunx Admin/AI/ML/BI/SP Consultant
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Worked with DevOps Engineering team in Revenue Digital Branch to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.

Frequently implemented AWS cloud-native design patterns. Such as highly scalable and fault-tolerant **Amazon DynamoDB Amazon SQS** to send messages between microservices, or to queue tasks for later processing. **SNS** to send messages to a large number of subscribers or to send notifications to customers, or to trigger workflows in other applications.

Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.

Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.

Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.
DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
Delivered and managed multi-cloud networks, enhancing operational efficiency and network reliability.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation and management of deployed machine learning models in production environments.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Implemented and maintained AKS clusters for various microservices applications using Terraform and Helm Charts.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Either using BMC Control-M v8.0 to manage, multiple data transformation jobs or batch scheduling via AutoSys
Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Principle DevOps Engineer /Sr. Data Engineer/Clouds/Linux Admin. /ML/SharePoint /K2 Developer/EDW
Responsibilities:
Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
Worked with DevOps Engineering team to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Implemented AWS cloud-native design patterns, including leveraging Amazon CloudWatch for comprehensive monitoring and observability of AWS resources. This includes tracking metrics, logs, and events from both applications & infrastructure. Additionally, utilized Amazon Cognito for robust user identity and access management in web and mobile applications.
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.
Leverage AWS services for cloud-native design patterns:
Highly scalable and fault-tolerant monitoring: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.

Unified user authentication and authorization: Implement Amazon Cognito for user identity and access management across web and mobile applications.

Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.

Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.

Docker Images: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Completed design documentation, data models, loading approaches and source to target mapping documents.

Leveraged technical expertise to consult on projects throughout their lifecycle. Conducted pre-project assessments to identify and mitigate risks, interdependencies, and exceptions. Developed solutions aligned with ICT strategy and architecture. Authored technical documentation to support new and existing applications.

Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevSecOps best practices to IT and business stakeholders.

Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.

Migrated on-premise applications to the cloud using AWS Direct Connect, Storage Gateway and AWS Snowball Edge.

Streamlined the DevOps workflow on AWS using Atlassian tools: Jira, Confluence, Bitbucket, Bamboo, PowerShell & Bash.

Executed implementation, documentation, integration, and support operations of a variety of REST API projects

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Quality Assurance Engineer/Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve software features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer/Assistant Quality Assurance Engineer/Software Test Engineer/SP Consultant
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture

- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Actively contribute to and guide innovation in DevOps practices and incorporate agile delivery approaches and technologies that assist with the development and delivery of high-quality software
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.

- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Machine Learning Engineer – Pathway Recruitment – 15/3/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior ML Engineer/Senior Data Engineer/Senior Data Analyst /Senior DevOps Engineer /Clouds/Linux Administrator /Python /R Developer /AI /ML /BI /SharePoint /Dynamics 365 CRM Consultant /EDW - Japan(7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Senior ML Engineer/Sr. DevOps/Data Engineer/Python/R Developer/AI/ML/BI/SP/D365 CRM Consultant/EDW
Responsibilities:

- Built a scalable Data Lake with Terraform for ingesting data from various sources, including potentially multiple clouds. This lake stores raw data and curated datasets for easy access. It seamlessly integrates with a multi-layered Snowflake analytics platform, allowing ready-to-use ML tools from AWS & Azure to power analytics, low-latency processing, and large-scale ML.
- Develop and deploy machine learning models using Generative AI, NLP, and LLMs to solve complex business problems.
- Developed and deployed machine learning models for image classification, object detection, and face recognition using frameworks like PyTorch and OpenCV. Additionally, implemented a Generative Adversarial Network (GAN) for an image-based AI system for cancer detection. This approach differs from Dr. Cao Kei's "PANDA" model for early pancreatic cancer detection, a deep learning-based system published in Nature Medicine in November 2023.
- Developed a secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Designed and implemented a Natural Language Processing (NLP) pipeline for sentiment analysis, resulting in a 7% improvement in customer satisfaction ratings.
- Collaborate with data scientists and engineers on the ML lifecycle: problem definition, data exploration, model selection (classification, regression, clustering), and implementation for scalability.

- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to
- **Build and train the ML model**: Leverage GCP tools (Vertex AI, Cloud Storage, Dataflow) to design the pipeline, split data for training and evaluation, and engineer features for optimal model performance.
- **Train and optimize the model**: Select the right type (XGBoost, TensorFlow etc.), train it on Vertex AI with hyperparameter tuning, and evaluate using metrics and visualizations on the validation set.
- **Deploy and monitor**: Use Vertex AI Endpoints for serving predictions, consider containers for flexibility. Monitor with Explainable AI and log predictions/errors for ongoing analysis.
- **Trained neural network**: A complex system inspired by the human brain, designed to learn by processing large amounts of data and identifying patterns. These patterns are then used to make predictions or classifications on new data.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Developed and maintained CI/CD pipelines for a variety of applications using Azure DevOps, incorporating DevSecOps practices to enhance security and move towards AIOps principles.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Develop medical-tailored LLMs (BERT, GPT-4, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Utilized AI/ML technologies like CrewAI and MLOps to automate tasks, improve productivity, and reduce costs in sales, customer service, and model building. Leveraged TinyML to integrate AI with IoT/mobile devices for innovative solutions.
- **Diffusion Models and LLMs for Image and Content Generation**: Using diffusion models like Stable Diffusion or Dall-E 2 to create new products, generate realistic avatars for the metaverse. Using LLMs for image& content generation by Sora etc.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Sr. ML Engineer/Sr. DevOps/Data Engineer /Python/R Developer/AI/ML/SharePoint BI Consultant /EDW
Responsibilities:

Built a scalable Data Lake with Terraform (IaC) for ingesting data from various sources, including potentially multiple clouds. This lake stores raw data and curated datasets for easy access. It seamlessly integrates with a multi-layered Snowflake analytics platform, allowing ready-to-use ML tools from both AWS and GCP to power analytics, low-latency processing, and large-scale machine learning within a hybrid cloud environment.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Experimented with novel algorithms and techniques, resulting in significant model performance improvements.

Leveraged automated techniques like hyperparameter tuning, model compression, and ensemble learning to optimize models in hybrid clouds. This empowered data scientists to automatically adjust a model's settings, reduce its size, or combine predictions for better accuracy without manual intervention.

Worked closely with cross-functional teams to integrate ML models into production systems.

Automated Fraud Detection: Developed and implemented a system that automatically trains and deploys a fraud detection model across cloud platforms. The system utilizes historical data from various sources and optimizes model accuracy for real-time transaction processing.

Scalable Image Classification: Designed and implemented a system for automated training, compression, and deployment of an image classification model in a hybrid cloud environment. This system ensures efficient resource utilization and leverages ensemble learning techniques (optional: to improve classification accuracy).

As a Data and Software Engineer, design and implement ETL pipelines either using GCP's Dataflow or AWS Glue or AWS EMR (with Apatche Spark /Hive, etc) to transform and load data from various sources, optimizing data processing efficiency.

Advanced Machine Learning Expertise: Developed and deployed complex ML models using Generative AI, NLP, and LLMs to solve critical business problems. Collaborated with data science teams to design and implement scalable solutions.
Machine Learning with R proficiency: Demonstrated proficiency in R for building classical machine learning models like logistic regression (loan default prediction) and linear regression (house price forecasting).
Utilized strong Python programming skills and expertise in ML libraries (TensorFlow, PyTorch) to build and deploy machine learning models for various applications.
Managed the entire ML lifecycle from data pre-processing, feature engineering, model training, evaluation to deployment.
Designed and implemented a **multi-tier Data Vault architecture** using Terraform to support data ingestion from multiple cloud providers and integrate with a Snowflake-based multi-layered analytics architecture.
Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.
Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.
Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
Programmatically interact with Power BI data and metadata using the Power BI REST API, accessible through R packages like `powerbi`. Embed Power BI reports and visuals in R Shiny apps using the Power BI Embedded API and dedicated R packages like `poweRbi` or `embedR`. Utilize RStudio Server on AWS instances for remote R development and execution.

- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Nvidia DGX-1 system

Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).
Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
Utilized distributed file systems like HDFS (Hadoop Distributed File System) or GCS (Google Cloud Storage) to efficiently store and access terabytes of data across multiple machines.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior ML Engineer/Sr. DevOps/Data Engineer /Python/R Developer/AI/ML/SharePoint /K2 Developer /EDW
Responsibilities:

- Designed and implemented **a multi-tier Data Vault architecture** using Terraform (IaC) to support data ingestion from various sources, potentially including multiple cloud providers. The architecture seamlessly integrates with a **Snowflake-based, multi-layered analytics platform**, empowering ready-to-use ML tools from both AWS and Azure to support analytics, low-latency processing, and machine learning use cases at scale.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Designed, built, and maintained scalable and robust data pipelines in Snowflake using Azure DevOps (or AWS CodePipeline) for CI/CD, DBT for data transformation, Alteryx for data preparation (optional), & Pydantic for data validation.
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Built custom machine learning models for personalized recommendations, achieving optimal performance through advanced feature engineering, A/B testing, and hyperparameter tuning. Also advised clients on ethical ML adoption and educated internal teams on Generative AI and NLP.
- Successfully collaborated with cross-functional teams to integrate ML models into existing workflows, streamlining processes and improving overall efficiency.
- Training Large Language Models (LLMs) through iteratively feeding them vast amounts of text data, generating predictions, and employing back propagation to adjust internal parameters, thereby progressively enhancing their language comprehension and generation capabilities.
- Neural networks use gradient descent to navigate a loss function landscape. The chain rule helps calculate the impact of each parameter (weight/bias) on the overall error, allowing gradient descent to adjust them iteratively, training the network.

- Managed SQL Server 2012/2016 development using SSIS, SSRS, SSAS and MDS for BI/DW initiatives.

Utilized either SSIS ETL capabilities or modern Power BI to extract complex data into a Data Warehouse (DW) staging area. Transformed and cleansed data was then loaded as Master Data to support various analytical needs. This includes Analysis Services Cubes (created with SSAS) for business reporting via SSRS or Data Marts for powering business applications.

Interactive Data Visualization: Capable of creating dashboards in Power BI by pinning various elements like tiles, slicers, and entire report pages. Experienced in utilizing other visualization tools like Tableau and QlikView for exploration purposes.

Advanced Data Analysis: Proficient in developing complex DAX formulas within Power BI to manipulate and analyze data for reports and visualizations.

Work closely with the development and product teams to bring their applications through the SDLC (software development lifecycle) from cloud-deployed development environments through to internal and protected production Kubernetes runtime environment using Kustomize, Helm and Ansible.

Build and support both Cloud development and on premises development/test and production environments.

Maintain and support the underlying Kubernetes platform (installed via Ansible and Kubespray with Cluster manager). Troubleshoot and resolve any issues that may occur with the Kubernetes runtime environment & applications running on it.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Designed data transformation workflows using either Azure Logic Apps or AWS Step Functions to orchestrate Spark Notebooks. Within these notebooks, leveraged Spark SQL or PySpark for data cleaning, enrichment, and aggregation tasks.

Engineer and productionise ML models for real-world impact, leveraging the Python, AWS stack and top-notch MLOps tools.

Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia

Health Sector: Machine Learning QA Specialist /Software Test Engineer/AI/ML /BI Developer /D365 CRM Consultant /DBA

Responsibilities:

- **Machine Learning QA Specialist**: Sole QA engineer within a dedicated ML squad, ensuring algorithm quality through manual and automated testing (language-agnostic scripting).
- **Collaborative Quality Assurance**: Works with cross-functional teams to understand system requirements, identify and document bugs, track issues, and perform regression testing after bug resolution.
- Developed robust ML models using synthetic data (Generative AI) and domain expertise. Optimized pipelines and ensured accuracy through precision-recall and Receiver Operating Characteristic (ROC) analysis.
- Seamless integration of Python code and libraries within R environments, enabling combined workflows.
- Proficient in designing and implementing data solutions on AWS, utilizing services like EC2, EBS, RDS, and S3, alongside expertise in Spark/PySpark SQL for big data querying and analysis, and Impala for low-latency requirements.
- Designed and built robust data pipelines, enabling efficient extraction from diverse sources and loading into data warehouses. Supported data quality initiatives, focusing on information security.
- Analyzed large-scale marketing campaign data to identify key performance indicators and optimize future campaigns.
- Utilised Azure ADF (Azure Data Factory) and Data Bricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, & deployment processes within Azure ecosystem.
- Utilized either SSIS ETL capabilities or modern Power BI to extract complex data into a Data Warehouse (DW) staging area. Transformed and cleansed data was then loaded as Master Data to support various analytical needs. This includes Analysis Services Cubes (created with SSAS) for business reporting via SSRS or Data Marts for powering business applications.
- Ensured robust container security involves leveraging technologies such as isolation, resource control, and access management, complemented by the implementation of container scanning tools like Snyk and Aqua Security using Scripting Languages: Python, Golang, Bash and PowerShell, including Azure Container Registry (ACR) or AWS ECR.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system
- Developed a comprehensive ad reporting tool using R's advanced statistical capabilities. The tool effectively analyzes complex data sets to deliver actionable insights for our client's upcoming product launch. Additionally, seamless integration with Dynamics 365 CRM facilitates efficient marketing workflows and follow-up activities.
- Utilized data modeling techniques like entity-relationship modeling and Data Vault to structure data for efficient querying and analysis. Develop, test, and optimize ETL/data objects with data modelers/analysts.

Developed automated CI/CD pipelines to accelerate development, testing, and release processes.

Spearheaded data governance initiatives, ensuring data integrity and trust.

- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet

- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Machine Learning QA Engineer/Software Test Engineer /Data Engineer /SharePoint Consultant /DBA
Responsibilities:

- **Software Quality Assurance & Automation**: Analyzes, documents, and tests software systems while ensuring quality and reliability. Develops and implements automated testing strategies, collaborating with cross-functional teams to stay current on best practices.
- **Advanced NLP & Machine Learning Expertise**: Explored cutting-edge NLP techniques, developed custom LLMs for chatbots, and published research on transfer learning approaches. Collaborated with data scientists and stakeholders to translate business needs into technical solutions using machine learning.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis.
- Developed and maintained SQL queries for data retrieval and manipulation.
- Performed data analysis and reporting to support business decision-making.
- Engineered end-to-end data pipelines to extract, transform, and load data from diverse sources to data warehouses.
- Utilized Jira and Confluence for effective project management and documentation.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Ops Engineer /Data Analyst /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Data Engineer /Data Analyst /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.

- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark SQL, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Data Engineer – GKE, EKS, AKS – TRC – Centre Group -Anne-Maria 10/3/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Data Engineer /Senior DevOps Engineer /Senior Data Analyst /Python/R Developer/Clouds/Linux Administrator /AI/ML/BI/SharePoint/Dynamics 365 CRM Consultant /EDW - Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Data Engineer/Senior DevOps Engineer /Python/R Developer /AI/ML /SP/D365 CRM Consultant /EDW
Responsibilities:

- Designed and implemented a **Google Cloud Platform EDW** using a serverless Lambda architecture. Data is ingested via Pub/Sub layer and processed in real-time streaming layer using Cloud Functions. Cloud Dataflow performs periodic batch processing to update the data warehouse stored in BigQuery under serving layer, as the single source of truth for analytics.
- Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Implemented a scalable data warehouse leveraging GCP's BigQuery service, facilitating efficient querying and analysis of extensive datasets through dimensional data modeling.

- **Big data wrangling**: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- Utilized distributed file systems like HDFS (Hadoop Distributed File System) or GCS (Google Cloud Storage) to efficiently store and access terabytes of data across multiple machines.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Developed a secure, local AI system for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision) for Imaging AI cancer detection, like pancreatic cancer, liver cancer..,
- Proficient in designing and implementing data solutions on AWS, utilizing services like EC2, EBS, RDS, and S3, alongside expertise in Spark/PySpark SQL for big data querying and analysis, and Impala for low-latency requirements.
- Leveraged SAP Data Services and SAP Business Objects to deliver robust ETL, Reporting, and Analytics solutions, showcasing in-depth understanding of these processes.
- Built robust & secure GCP Landing Zones with Terraform, ensuring compliance with HIPAA, GDPR, and PCI DSS standards, and cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
- Developed and maintained CI/CD pipelines for a variety of applications using GCP DevOps, DevSecOps towards AIOps.
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, ANN analytics, etc.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Utilized Cloud Composer or Airflow for orchestrating workflows and advanced scheduling within GCP Dataflow. These tools integrate well with Dataflow and offer a native GCP experience.
- Employ Dataform for data pipeline definition as code, managing SQL code with version control, data asset definition, and pipeline documentation. Utilized Dataform to manage the "what" of the pipelines, while Cloud Composer/Airflow handles the "when" and "how" of execution. Used agile methodologies to deliver projects on time and within budget.
- Deploying GKE as a managed Kubernetes service for running containerized applications on GCP, managing Kubernetes clusters with automatic scaling, load balancing, and health checks to ensure high availability and scalability of applications.
- Developed and maintained IaC scripts using Python and Terraform for provisioning and managing cloud resources, enhancing scalability and flexibility of applications. Developing Data APIs (REST, GraphQL, Graph API, gRPC)

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior Data Engineer/Senior DevOps Engineer /Python/R Developer /AI/ML /BI/SP Consultant /EDW
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Researched and implemented appropriate ML algorithms and tools based on business requirements.

Utilize Cloud Composer or Airflow for orchestrating workflows and advanced scheduling within GCP Dataflow.

Configure dbt for BigQuery by specifying the project ID, dataset, and other connection details in the profiles.yml file

Automated data pipelines using Airflow and Dataform, ensuring efficient data processing and delivery.

Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.

Collaborated with business analysts and product owners to translate data requirements into actionable solutions, driving data-driven decision-making. Utilized Talend for ETL/ELT processes, achieving streamlined data integration.

Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.

Leveraged Cloud Build to automate building and testing of data pipeline code. Utilized Cloud Storage to securely store data pipeline code and artifacts. Cloud Monitoring tracked the performance of data pipeline

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.

Implement automated monitoring with Prometheus and Grafana to collect metrics for all RBA's data pipelines.

Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.

Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to

Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Seamlessly execute automated tests within GCP utilizing Selenium, Cypress, or Appium. Leverage GCP's scalable infrastructure like Compute Engine or Kubernetes Engine for parallel test execution, ensuring effortless adaptation to growing business needs.

Conduct non-functional acceptance testing, develop test scripts, and execute test cases for system enhancements.

Leveraged SAP Data Services and SAP Business Objects to deliver robust ETL, Reporting, and Analytics solutions, showcasing in-depth understanding of these processes.

Migrated on-premise applications to the Google Cloud Platform (GCP) cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos, and CloudEndure Migration to GKE environment.

Employing Dataform for data pipeline orchestration, SQL code management with version control, data asset definition, testing and deployment environment creation, and pipeline documentation.

Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.

Deploy and manage microservices-based applications using Docker CRE and Kubernetes on GCP GKE clusters.

Administered and monitored both Windows and Linux-based systems, ensuring smooth operation of critical applications.

Integrate security, code analysis, and quality checks into the SDLC automation using SonarQube, Snyk, and OWASP ZAP.

Monitor and optimize system and application performance using Stackdriver, Datadog, Splunk and New Relic.

Engineered AI-driven ML systems using Python and machine learning frameworks (PyTorch, scikit-learn)

Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)

Gov. University: Senior Data Engineer/Senior DevOps Engineer/Python/R Developer /AI/ML/SharePoint /K2 Developer/DBA

Responsibilities:

Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.

Leverage Kubespray with its community-maintained extensions and Ansible to automate Kubernetes cluster deployment on AWS and Azure clouds. Then, utilize Kustomize for managing Kubernetes applications deployed on Amazon EKS and Azure AKS, enabling consistent configurations and simplified deployments across both cloud providers.

Migrated on-premises data platforms to GCP, leveraging services like BigQuery, Dataflow, and Cloud Storage.

Engineered and operated data platforms on Google Cloud Platform, utilizing services such as BigQuery, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.

Designing and developing complex data pipelines in a data analytics environment, dealing with large datasets

Utilize Cloud Composer or Airflow for orchestrating workflows and advanced scheduling within GCP Dataflow.

Configure dbt for BigQuery by specifying the project ID, dataset, and other connection details in the profiles.yml file

Automated data pipelines using Airflow and Dataform, ensuring efficient data processing and delivery.

Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.

Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to

As a product development team member, utilized tools like Selenium, Cypress, and Appium to run automated tests seamlessly within GCP. Leveraging GCP's scalable infrastructure like Compute Engine or Kubernetes Engine ensures parallel test execution and effortless adaptation to growing business needs.

Proficient in designing and implementing data solutions on AWS, utilizing services like EC2 for data compute, EBS for storage, RDS for databases, and S3 for data lakes, alongside expertise in Spark/PySpark SQL for big data querying and analysis, and Impala for low-latency analytical workloads.

Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.

Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Leveraging Dataform for managing intricate data pipelines with dependency management, data quality checks, and data lineage tracking. Using Cloud Build to automate diverse CI/CD tasks such as building, testing, and deploying applications to various platforms like Kubernetes, Cloud Functions, and App Engine.

Leveraging BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment.

Conducted non-functional acceptance testing, developed test scripts, and executed non-functional test cases for Data and Integration systems including the deployment of application and data pipeline releases while ensuring operational process and procedural documentation are maintained professionally.

Completed design, data models, loading approaches and source to target mapping documents.

Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.

Created a mapping document that defines how the source data fields correspond to the target data fields.

Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.

Implemented data transformation using Cloud Workflows and Dataproc Notebooks, writing BigQuery SQL scripts or utilizing PySpark for transformations, including data cleaning, enrichment, and aggregation.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,

Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases

- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Data Engineer /DevOps Engineer/Clouds/Linux Administrator/SharePoint Consultant

Responsibilities:

- Implemented a **3-tier lambda architecture EDW on Azure, combining batch and real-time processing**. An **ingestion layer** handles data intake, followed by a **batch processing layer** using HDInsight and PySpark for scalable data transformation. The **real-time analysis layer** leverages Azure Stream Analytics on Data Streams for near real-time insights. Additionally, Databricks SQL offers interactive querying on Data Lake Storage data.
- Migrated on-premises data platforms to GCP, leveraging services like BigQuery, Dataflow, and Cloud Storage.
- Deploying GKE as a managed Kubernetes service for running containerized applications on GCP, managing Kubernetes clusters with automatic scaling, load balancing, and health checks to ensure high availability and scalability of applications.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all UC's data pipelines.
- Leveraging BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment.
- Created and implemented ETL processes to extract, transform, and load data into data warehouses and data lakes.
- Implement the infrastructure as code using Terraform and Google Cloud Deployment Manager.
- Implemented secure CI/CD pipelines using Terraform, Vault, Artifactory, and GitLab for various applications.
- Implemented scripting solutions (Python, Bash) to automate routine tasks and improve operational efficiency.
- Provided technical support to internal users and troubleshoot data access and quality problems.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Demonstrated proficiency in scripting languages such as Python/PowerShell, Ansible, Puppet and Go for automation.
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery

- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Seamlessly execute automated tests within GCP utilizing Selenium, Cypress, or Appium/Appgee. Leverage GCP's scalable infrastructure like Compute Engine or Kubernetes Engine for parallel test execution, ensuring effortless adaptation to growing business needs. Monitored and maintained system performance, identified and resolved performance bottlenecks.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: **Data Engineer /DevOps Engineer /ML /SharePoint Consultant**
Responsibilities:

- Performed data analysis using SQL and reporting tools to support business intelligence initiatives.
- Developed and maintained data reports and dashboards for various stakeholders.
- Cleaned and transformed raw data to ensure data quality and consistency.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Shell, Ansible and Puppet
- Proficient in scripting and programming languages, including Python and Ruby.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Configured and managed DNS, DHCP, Active Directory, Exchange Server, SharePoint Server, SQL Server, and
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: **Data Ops Engineer /Data Analyst /BI Developer /Senior Systems Engineer /DWH /DBA**
Responsibilities:

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: **Project Manager /Data Engineer /Data Analyst /BI Consultant /DWH**
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

AI & Data Engineer -Ethos BeathChapman – 7/Mar/24
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI & Data Engineer/Senior Data Analyst /Senior DevOps Engineer /Clouds/Linux Administrator /Python /R Developer AI /ML /BI /SharePoint /Dynamics 365 CRM Consultant /EDW -Japan(7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI & Data Engineer/Sr. DevOps Engineer/Python/R Developer/AI/ML/BI/SP/D365 CRM Consultant/EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Designed and implemented **a multi-tier Data Vault architecture** using Terraform (IaC) to support data ingestion from various sources, potentially including multiple cloud providers. The architecture seamlessly integrates with a

Snowflake-based, multi-layered analytics platform, empowering ready-to-use ML tools from both AWS and Azure to support analytics, low-latency processing, and machine learning use cases at scale.

- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Possess expertise in utilizing various cloud platforms (Azure, AWS, and Snowflake) to build automated data pipelines, including data ingestion (using tools like ADF pipelines, Glue Data Pipelines, or Snowpipes), change data capture (using Synapse Analytics CDC, Kinesis Data Streams, or Snowflake Streams), and task orchestration (using ADF, Databricks notebooks, Step Functions, or Glue workflows).
- Utilized Snowflake Monitoring and Snowflake Query Profile to monitor and optimize Snowflake data platform performance.
- **Cloud-based, modular data platform**: Leverage scalable cloud resources (AWS or Azure) to build a flexible platform supporting diverse workloads (analytics, ML, etc.). Utilize tools like Snowflake (data warehousing), Azure Data Factory/AWS Glue (data integration), and CI/CD pipelines for secure and automated data delivery.
- Leveraged Microsoft Power Apps to design user interfaces and workflows for data management. Utilized Power Automate for automating data-related tasks, such as data collection, manipulation, and integration with various data sources.
- **DataOps best practices**: Implement data governance, information security, and quality control throughout the data lifecycle. Foster collaboration across teams to ensure data standards and integration across applications and infrastructure.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Managed SQL Server 2019/2022 development using SSIS, SSRS, SSAS, DAX Studio and MDS for BI/DW initiatives.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio, which can be securely searched by PrivatGPT2.0 or ChatGPT Enterprise
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Implemented a secure, local version of CrewAI that does not require an internet connection. The system uses versatile AI agents to assist Business Analysts and Data Analysts in efficient data analysis.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Developed and maintained CI/CD pipelines for a variety of applications using Azure DevOps, incorporating DevSecOps practices to enhance security and move towards AIOps principles.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision) for Imaging AI cancer detection, like pancreatic cancer, liver cancer...
- Expertise in designing and implementing data solutions on AWS using services like EC2, EBS, RDS, and S3. Proficient in utilizing Apache Spark for big data querying (Hive tables, Spark SQL/PySpark SQL) and analysis, along with Impala for low-latency needs
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Sr. AI & Data Engineer /Sr. DevOps Engineer/Python/R Developer/AI/ML/SharePoint BI Consultant /EDW
Responsibilities:

Designed and implemented a **Google Cloud Platform EDW** using a serverless Lambda architecture. Data is ingested via Pub/Sub and processed in real-time using Cloud Functions. Cloud Dataflow performs periodic batch processing to update the data warehouse stored in BigQuery, which serves as the single source of truth for analytics.

Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Designed and implemented a **multi-tier Data Vault architecture** using Terraform to support data ingestion from multiple cloud providers and integrate with a Snowflake-based multi-layered analytics architecture.

Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.
Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.
Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
Utilized Azure DevOps, Azure Data Factory, and Power automate to streamline data workflows and automation processes.
Managed SQL Server 2016/2017 development using SSIS, SSRS, SSAS and MDS for BI/DW initiatives.
Designed, built, and maintained data processing systems for migrating and refactoring ETL pipelines in Cloudera, enabling hybrid cloud deployments. Additionally, built ETL data pipelines for migration purposes.
Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
Designed & deployed a modern **AWS Lambda Architecture EDW**, comprising: **Batch layer**: Hadoop HDFS for distributed storage and MapReduce for batch processing of large datasets. **Speed layer**: Spark for in-memory processing of real-time data streams, utilizing RDDs for efficient data manipulation. **Service layer**: Spark Streaming for low-latency analytics on streaming data, generating SNS application notifications based on real-time insights. **Direct write queries**: Spark SQL for efficiently querying datasets within Spark cluster, enabling seamless integration with downstream applications & services.
Design, build, and maintain scalable and robust data pipelines in Snowflake using Azure DevOps (or AWS CodePipeline) for CI/CD, DBT for data transformation, Alteryx for data preparation (optional), and Pydantic for data validation.
Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Designed and implemented a data processing flow that integrates Data Lake Gen2.0 (or Data Lake EDW), Databricks, and Synapse Analytics to enable efficient data transformation and analysis:
Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity
Programmatically interact with Power BI data and metadata using the Power BI REST API, accessible through R packages like `powerbi`. Embed Power BI reports and visuals in R Shiny apps using the Power BI Embedded API and dedicated R packages like `poweRbi` or `embedR`. Utilize RStudio Server on AWS instances for remote R development and execution.

- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Unleash insights from diverse data: apply unsupervised ML to uncover patterns in unlabeled data, leverage RStudio for supervised learning, and expertly select algorithms (e.g., KNN, CNN, RNN) aided by AutoML tools like Amazon SageMaker Autopilot, Azure AutoML, H2O.ai, TPOT and KerasTuner.
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Nvidia DGX-1 system

Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Machine Learning with R for logistic regression user case, linear regression with user case
Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DataOps Engineer/Sr. DevOps Engineer/Python/R Developer/AI/ML/SharePoint /K2 Developer /EDW
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Managed SQL Server 2012/2016 development using SSIS, SSRS, SSAS and MDS for BI/DW initiatives.

Utilized either SSIS ETL capabilities or modern Power BI to extract complex data into a Data Warehouse (DW) staging area. Transformed and cleansed data was then loaded as Master Data to support various analytical needs. This includes Analysis Services Cubes (created with SSAS) for business reporting via SSRS or Data Marts for powering business applications.

Orchestrated end-to-end data processing workflows. Collaborated with cross-functional teams to enhance data consumption patterns. Developed infrastructure patterns for efficient data processing. Ensured confidentiality, integrity, and availability of sensitive information. Utilized SSIS and SSRS for SQL Server 2012/2016 development.

Interactive Data Visualization: Capable of creating dashboards in Power BI by pinning various elements like tiles, slicers, and entire report pages. Experienced in utilizing other visualization tools like Tableau and QlikView for exploration purposes.

Advanced Data Analysis: Proficient in developing complex DAX formulas within Power BI to manipulate and analyze data for reports and visualizations.

Work closely with the development and product teams to bring their applications through the SDLC (software development lifecycle) from cloud-deployed development environments through to internal and protected production Kubernetes runtime environment using Kustomize, Helm and Ansible.

Build and support both Cloud development and on prem development/test and production environments.

Maintain and support the underlying Kubernetes platform (installed via Ansible and Kubespray with Cluster manager). Troubleshoot and resolve any issues that may occur with the Kubernetes runtime environment & applications running on it.

Work closely with the Unix, Networks and Security teams to ensure that all our platforms are running in an optimal and secure fashion. Perform and automate release and deployment management (fitting in with the Service Management strategy). Proficient in the use of, scritping (e.g. bash/python) and administration aspects of linux/unix systems.

Support the development tool sets that the team look after: Jenkins, Nexus, Atlassian Jira/Confluence/bitbucket, Chartmuseum, GitLab.

Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Using either BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or batch scheduling via AutoSys.

Designed data models leveraging Kimball and Data Vault methodologies for efficient data retrieval.

Designed data transformation workflows using either Azure Logic Apps or AWS Step Functions to orchestrate Spark Notebooks. Within these notebooks, leveraged Spark SQL or PySpark for data cleaning, enrichment, and aggregation tasks.

Created interactive Power BI dashboards with DAX calculations, utilizing Databricks for complex analytics.

Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: DataOps Engineer/Data Analyst /DevOps Engineer/SharePoint /BI Developer /D365 CRM Consultant /DBA
Responsibilities:

- Designed and built robust data pipelines, enabling efficient extraction from diverse sources and loading into data warehouses. Supported data quality initiatives, focusing on information security.
- Collaborated with DevOps teams to integrate data from various sources into a high-quality information repository.
- **Docker Images:** Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Seamless integration of Python code and libraries within R environments, enabling combined workflows.
- Analyzed large-scale marketing campaign data to identify key performance indicators and optimize future campaigns.
- Utilised Azure ADF (Azure Data Factory) and Data Bricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, & deployment processes within the Azure ecosystem.
- Utilized either SSIS ETL capabilities or modern Power BI to extract complex data into a Data Warehouse (DW) staging area. Transformed and cleansed data was then loaded as Master Data to support various analytical needs. This includes Analysis Services Cubes (created with SSAS) for business reporting via SSRS or Data Marts for powering business applications.
- Ensured robust container security involves leveraging technologies such as isolation, resource control, and access management, complemented by the implementation of container scanning tools like Snyk and Aqua Security using Scripting Languages: Python, Golang, Bash and PowerShell, including Azure Container Registry (ACR) or AWS ECR.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system
- Developed a comprehensive ad reporting tool using R's advanced statistical capabilities. The tool effectively analyzes complex data sets to deliver actionable insights for our client's upcoming product launch. Additionally, seamless integration with Dynamics 365 CRM facilitates efficient marketing workflows and follow-up activities.

- Utilized data modeling techniques like entity-relationship modeling and Data Vault to structure data for efficient querying and analysis. Develop, test, and optimize ETL/data objects with data modelers/analysts.
- Automate with Ansible any infrastructure that the platform team install and configure.
- Collaborate with teams to integrate datasets within Azure Cloud (Databricks). Analyze metadata to ensure data quality and consistency. Provide insights and recommendations for metadata management best practices.

Develop and maintain documentation related to metadata processes and configurations
Developed automated CI/CD pipelines to accelerate development, testing, and release processes.
Spearheaded data governance initiatives, ensuring data integrity and trust.

- Defined and enforced data architecture standards across applications and infrastructure.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DataOps Engineer /Data Analyst /DevOps Engineer/SharePoint Consultant /DBA
Responsibilities:

- Developed and maintained automated CI/CD pipelines for efficient and secure data delivery.
- Implemented best practices for data governance and information security to ensure data integrity and compliance.
- Developed and maintained data pipelines for ingesting and processing data from various internal and external sources, including APIs, databases, and flat files.
- Collaborated with data scientists and business stakeholders to translate business needs into technical solutions.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Developed and maintained SQL queries for data retrieval and manipulation.
- Performed data analysis and reporting to support business decision-making.
- Engineered end-to-end data pipelines to extract, transform, and load data from diverse sources to data warehouses.
- Utilized Jira and Confluence for effective project management and documentation.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Ops Engineer /Data Analyst /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Data Engineer /Data Analyst /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.

- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark SQL, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DataOps Engineer – MA Financial – The Argyle Network – 6/Mar/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DataOps Engineer/Senior Data Analyst /Senior DevOps Engineer /Clouds/Linux Administrator /Python /R Developer AI /ML /BI /SharePoint /Dynamics 365 CRM Consultant /EDW -Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DataOps Engineer/Sr. DevOps Engineer/Python/R Developer/AI/ML/BI/SP/D365 CRM Consultant/EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Designed and implemented **a multi-tier Data Vault architecture** using Terraform (IaC) to support data ingestion from various sources, potentially including multiple cloud providers. The architecture seamlessly integrates with a **Snowflake-based, multi-layered analytics platform**, empowering ready-to-use ML tools from both AWS and Azure to support analytics, low-latency processing, and machine learning use cases at scale.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Possess expertise in utilizing various cloud platforms (Azure, AWS, and Snowflake) to build automated data pipelines, including data ingestion (using tools like ADF pipelines, Glue Data Pipelines, or Snowpipes), change data capture (using Synapse Analytics CDC, Kinesis Data Streams, or Snowflake Streams), and task orchestration (using ADF, Databricks notebooks, Step Functions, or Glue workflows).
- Utilized Snowflake Monitoring and Snowflake Query Profile to monitor and optimize Snowflake data platform performance.
- Assessed and adopted new cloud technologies, creating production-ready usage patterns.
- **Cloud-based, modular data platform**: Leverage scalable cloud resources (AWS or Azure) to build a flexible platform supporting diverse workloads (analytics, ML, etc.). Utilize tools like Snowflake (data warehousing), Azure Data Factory/AWS Glue (data integration), and CI/CD pipelines for secure and automated data delivery.
- Leveraged Microsoft Power Apps to design user interfaces and workflows for data management. Utilized Power Automate for automating data-related tasks, such as data collection, manipulation, and integration with various data sources.
- **DataOps best practices**: Implement data governance, information security, and quality control throughout the data lifecycle. Foster collaboration across teams to ensure data standards and integration across applications and infrastructure.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Managed SQL Server 2019/2022 development using SSIS, SSRS, SSAS, DAX Studio and MDS for BI/DW initiatives.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, with a focus on M365 environments (O365, MS Teams, SharePoint 2019, Dynamics 365 CRM, Power BI, Power Automate, Power Apps etc.).
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Developed and maintained CI/CD pipelines for a variety of applications using Azure DevOps, incorporating DevSecOps practices to enhance security and move towards AIOps principles.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision) for Imaging AI cancer detection, like pancreatic cancer, liver cancer...
- Expertise in designing and implementing data solutions on AWS using services like EC2, EBS, RDS, and S3. Proficient in utilizing Apache Spark for big data querying (Hive tables, Spark SQL/PySpark SQL) and analysis, along with Impala for low-latency needs
- Designed data transformation workflows using either Azure Logic Apps or AWS Step Functions to orchestrate Spark Notebooks. Within these notebooks, leveraged Spark SQL or PySpark for data cleaning, enrichment, and aggregation tasks.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Sr. DataOps Engineer/Sr. DevOps Engineer/Python/R Developer/AI/ML/SharePoint BI Consultant /EDW
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Researched and implemented appropriate ML algorithms and tools based on business requirements.
Designed and implemented a multi-tier Data Vault architecture using Terraform to support data ingestion from multiple cloud providers and integrate with a Snowflake-based multi-layered analytics architecture.
Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
Utilized Azure DevOps, Azure Data Factory, and Power automate to streamline data workflows and automation processes.
Managed SQL Server 2016/2017 development using SSIS, SSRS, SSAS and MDS for BI/DW initiatives.
Designed, built, and maintained data processing systems for migrating and refactoring ETL pipelines in Cloudera, enabling hybrid cloud deployments. Additionally, built ETL data pipelines for migration purposes.
Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
Designed & deployed a modern **AWS Lambda Architecture EDW**, comprising: **Batch layer**: Hadoop HDFS for distributed storage and MapReduce for batch processing of large datasets. **Speed layer**: Spark for in-memory processing of real-time data streams, utilizing RDDs for efficient data manipulation. **Service layer**: Spark Streaming for low-latency analytics on streaming data, generating SNS application notifications based on real-time insights. **Direct write queries**: Spark SQL for efficiently querying datasets within Spark cluster, enabling seamless integration with downstream applications & services.
Design, build, and maintain scalable and robust data pipelines in Snowflake using Azure DevOps (or AWS CodePipeline) for CI/CD, DBT for data transformation, Alteryx for data preparation (optional), and Pydantic for data validation.
Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity
Collaborated with IT and business stakeholders to integrate analytics and data science outputs into business processes.
Programmatically interact with Power BI data and metadata using the Power BI REST API, accessible through R packages like `powerbi`. Embed Power BI reports and visuals in R Shiny apps using the Power BI Embedded API and dedicated R packages like `poweRbi` or `embedR`. Utilize RStudio Server on AWS instances for remote R development and execution.

- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Executed implementation, documentation, integration, and support operations of a variety of REST API projects
- Unleash insights from diverse data: apply unsupervised ML to uncover patterns in unlabeled data, leverage RStudio for supervised learning, and expertly select algorithms (e.g., KNN, CNN, RNN) aided by AutoML tools like Amazon SageMaker Autopilot, Azure AutoML, H2O.ai, TPOT and KerasTuner.
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- **Automation and CI/CD**: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure. Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Nvidia DGX-1 system

Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Utilized data modeling techniques like entity-relationship modeling and Data Vault to structure data for efficient querying and analysis. Implemented data warehousing solutions using platforms like Amazon Redshift and Snowflake.
Machine Learning with R for logistic regression user case, linear regression with user case
Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DataOps Engineer/Sr. DevOps Engineer/Python/R Developer/AI/ML/SharePoint /K2 Developer /EDW
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Managed SQL Server 2012/2016 development using SSIS, SSRS, SSAS and MDS for BI/DW initiatives.

Utilized either SSIS ETL capabilities or modern Power BI to extract complex data into a Data Warehouse (DW) staging area. Transformed and cleansed data was then loaded as Master Data to support various analytical needs. This includes Analysis Services Cubes (created with SSAS) for business reporting via SSRS or Data Marts for powering business applications.

Orchestrated end-to-end data processing workflows. Collaborated with cross-functional teams to enhance data consumption patterns. Developed infrastructure patterns for efficient data processing. Ensured confidentiality, integrity, and availability of sensitive information. Utilized SSIS and SSRS for SQL Server 2012/2016 development.

Interactive Data Visualization: Capable of creating dashboards in Power BI by pinning various elements like tiles, slicers, and entire report pages. Experienced in utilizing other visualization tools like Tableau and QlikView for exploration purposes.

Advanced Data Analysis: Proficient in developing complex DAX formulas within Power BI to manipulate and analyze data for reports and visualizations.

Work closely with the development and product teams to bring their applications through the SDLC (software development lifecycle) from cloud-deployed development environments through to internal and protected production Kubernetes runtime environment using Kustomize, Helm and Ansible.

Build and support both Cloud development and on prem development/test and production environments.

Maintain and support the underlying Kubernetes platform (installed via Ansible and Kubespray with Cluster manager). Troubleshoot and resolve any issues that may occur with the Kubernetes runtime environment & applications running on it.

Work closely with the Unix, Networks and Security teams to ensure that all our platforms are running in an optimal and secure fashion. Perform and automate release and deployment management (fitting in with the Service Management strategy). Proficient in the use of, scritping (e.g. bash/python) and administration aspects of linux/unix systems.

Support the development tool sets that the team look after: Jenkins, Nexus, Atlassian Jira/Confluence/bitbucket, Chartmuseum, GitLab.

Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Using either BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or batch scheduling via AutoSys.

Designed data models with Kimball or Data Vault methodologies for efficient data retrieval.

Created interactive Power BI dashboards with DAX calculations, utilizing Databricks for complex analytics.

Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Health Sector: DataOps Engineer/Data Analyst /DevOps Engineer/SharePoint /BI Developer /D365 CRM Consultant /DBA
Responsibilities:

- Designed and built robust data pipelines, enabling efficient extraction from diverse sources and loading into data warehouses. Supported data quality initiatives, focusing on information security.
- Collaborated with DevOps teams to integrate data from various sources into a high-quality information repository.
- **Docker Images:** Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Seamless integration of Python code and libraries within R environments, enabling combined workflows.
- Analyzed large-scale marketing campaign data to identify key performance indicators and optimize future campaigns.
- Utilised Azure ADF (Azure Data Factory) and Data Bricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, & deployment processes within the Azure ecosystem.
- Utilized either SSIS ETL capabilities or modern Power BI to extract complex data into a Data Warehouse (DW) staging area. Transformed and cleansed data was then loaded as Master Data to support various analytical needs. This includes Analysis Services Cubes (created with SSAS) for business reporting via SSRS or Data Marts for powering business applications.

- Ensured robust container security involves leveraging technologies such as isolation, resource control, and access management, complemented by the implementation of container scanning tools like Snyk and Aqua Security using Scripting Languages: Python, Golang, Bash and PowerShell, including Azure Container Registry (ACR) or AWS ECR.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system
- Developed a comprehensive ad reporting tool using R's advanced statistical capabilities. The tool effectively analyzes complex data sets to deliver actionable insights for our client's upcoming product launch. Additionally, seamless integration with Dynamics 365 CRM facilitates efficient marketing workflows and follow-up activities.
- Utilized data modeling techniques like entity-relationship modeling and Data Vault to structure data for efficient querying and analysis. Develop, test, and optimize ETL/data objects with data modelers/analysts.
- Automate with Ansible any infrastructure that the platform team install and configure.
- Collaborate with teams to integrate datasets within Azure Cloud (Databricks). Analyze metadata to ensure data quality and consistency. Provide insights and recommendations for metadata management best practices.

Develop and maintain documentation related to metadata processes and configurations
Developed automated CI/CD pipelines to accelerate development, testing, and release processes.
Spearheaded data governance initiatives, ensuring data integrity and trust.

- Defined and enforced data architecture standards across applications and infrastructure.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DataOps Engineer /Data Analyst /DevOps Engineer/SharePoint Consultant /DBA
Responsibilities:

- Developed and maintained automated CI/CD pipelines for efficient and secure data delivery.
- Implemented best practices for data governance and information security to ensure data integrity and compliance.
- Developed and maintained data pipelines for ingesting and processing data from various internal and external sources, including APIs, databases, and flat files.
- Collaborated with data scientists and business stakeholders to translate business needs into technical solutions.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Developed and maintained SQL queries for data retrieval and manipulation.
- Performed data analysis and reporting to support business decision-making.
- Engineered end-to-end data pipelines to extract, transform, and load data from diverse sources to data warehouses.
- Utilized Jira and Confluence for effective project management and documentation.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Ops Engineer /Data Analyst /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Data Engineer /Data Analyst /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.

- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark SQL, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Azure DevOps Engineer – Recruitment Hive – 4/Mar/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Azure DevOps Engineer /Senior Data Engineer /Clouds/Linux Administrator /Python /R Developer /AI/ML/DL/BI/SharePoint/Dynamics 365 CRM Consultant /EDW/DBA -Japan(7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Azure DevOps Engineer/Sr. Data Engineer/Python/R Developer/AI/ML/SP/D365 CRM Consultant/EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
- Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- **Architected and implemented a modern 3-layer lambda architecture on Azure**: **Batch layer**: Utilized Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes.
- **Speed layer**: Employed Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics.
- Designed and implemented automated build, test, and deployment pipelines using Azure DevOps, YAML, ARM templates, and PowerShell for 5+ companies.
- Migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. Utilized Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration. Additionally, Azure Migrate and Azure Site Recovery ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 18c/19c, and Access Manager for Oracle-deployed applications.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- **Developed and Documented Automation Technologies**: Defined, developed, and documented automation technologies to enhance the system's build, test, and deployment capabilities.
- Leveraged Azure DevOps, YAML, ARM Templates, and PowerShell to create efficient and reliable pipelines.
- **Continuous Integration (CI) and Continuous Deployment (CD)**: Collaborated with application development teams to maintain and enhance DevOps CI/CD pipelines.
- Orchestrated the entire software development lifecycle, from code commits to production deployment.
- **Project Cloud Migration and Solution Platform**: Actively participated in project cloud migration efforts. Guided the team in adopting best practices for Azure pipelines. Ensured smooth transitions between production systems &planned deployments.
- **Infrastructure as Code (IaC) Solutions**:
- Designed and implemented IaC solutions using Terraform as the primary IaC tool. Possess experience with defining infrastructure foundations for systems or Azure products using tools like ARM templates or LAB3 Bedrock, a pre-built IaC solution leveraging Terraform for faster cloud adoption on Azure based on the Microsoft Cloud Adoption Framework (CAF).
- **Microsoft Azure Services Optimization**: Configured, monitored, and optimized services on the Microsoft Azure platform.
- Ensured high availability, scalability, and security for critical applications.
- Defined, developed, and documented automation technologies to enhance build, test, and deployment capabilities for key projects in a large federal government program focused on agriculture.
- Implemented and maintained Azure Kubernetes Service (AKS) clusters for various microservices applications using Terraform and Helm. Automated tasks and scripts using Python, PowerShell and Bash
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, GitHub Actions, and Jenkins
- Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Executing training and retraining systems as needed, utilizing either traditional machine learning libraries like Scikit-learn, CNN, or RNN, or leveraging AutoML toolkits [e.g., Google Cloud AutoML, AWS SageMaker, Azure Machine Learning(AML)] for automated feature engineering, model selection, and hyperparameter tuning.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**

Gov. Central Bank: Senior Azure DevOps Engineer/Sr. Data Engineer/Python/R Developer/AI/ML/SP/BI Consultant /EDW
Responsibilities:

- Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Designing and maintaining DevOps CI/CD pipelines to streamline development processes.
- Guided project cloud migration efforts and further developed Azure pipelines to support the solution platform.
- Successfully migrated on-premises workloads to Azure Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Azure SQL. This involved leveraging Power BI Pro Get Data connectors for initial data access, followed by the Azure Database Migration Service for comprehensive database migration. Additionally, Azure Migrate and Azure Site Recovery ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 17/18c, and Access Manager for Oracle-deployed applications.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Maintained version control visibility between production systems & planned deployments to ensure consistency & reliability.
- Defined and documented existing automation technologies for improved system efficiency and build/deployment.
- Participated in cloud migration initiatives, leading the configuration & development of Azure pipelines for solution platforms.
- Maintained clear visibility of version control between production systems and planned deployments for enhanced release management using Plastic SCM for precise version control. Achieved consistency and reproducibility across environments.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Monitored and troubleshooted Azure services using Splunk, Data Dog and Azure Network Monitor
- Collaborated with application development teams to develop and maintain DevOps CI/CD pipelines, ensuring seamless integration and delivery. Codified infrastructure setup, including virtual machines, networking, and storage.
- Managed Linux servers using Ansible, Chef and Puppet; Used Maven, ANT and Gradle for build tools
- Expert in Docker and Compose, including multi-stage builds and advanced networking. Manage S3 efficiently, including bucket configuration, security best practices, and cost-optimization through batch operations. Secure cloud infrastructure by implementing best practices in access control, identity and access management (IAM), and vulnerability management.
- Configured and managed Azure resources such as virtual machines, storage accounts, app services, load balancers, and firewalls. Utilized Gitlab, GitHub and Bitbucket for source code management and version control
- High proficiency in Active Directory, Azure AD, Managed the installation, configuration, and testing of new and existing hardware, software, and WAN/LAN technologies using (IaC) tools: Azure Resource Manager(ARM) templates, Teraform or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code
- Streamlined the DevOps workflow on AWS using Atlassian tools (Jira, Confluence, Bitbucket, Bamboo), resulting in comprehensive lifecycle management.
- Executed implementation, documentation, integration, and support operations of a variety of REST API projects
- Unleash insights from diverse data: apply unsupervised ML to uncover patterns in unlabeled data, leverage RStudio for supervised learning, and expertly select algorithms (e.g., KNN, CNN, RNN) aided by AutoML tools like Amazon SageMaker Autopilot, Azure AutoML, H2O.ai, TPOT and KerasTuner.
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- **Automation and CI/CD**: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure. Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.
- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Nvidia DGX-1 system

Oct. 2013 – Nov.2015 University of Canberra (UC), ACT/SYD (Working remotely from Sydney Home Office)
Gov. University: Senior Azure DevOps Engineer/Sr. Data Engineer/Python/R Developer/AI/ML/SP/K2 Developer
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus

Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Implemented security best practices for Azure resources using Azure AD, Azure Key Vault and Azure Policies

Used Terraform and CloudFormation to write infrastructure as code for Azure resources

Collaborated with development teams to implement automated build and deployment pipelines using Azure DevOps, YAML, ARM Templates, and PowerShell.

Defined Infrastructure as Code (IaC) solutions using Terraform and incorporated them into automated pipelines to facilitate consistent and repeatable deployments.

Developed applications in Java, Microsoft .NET, and Oracle-deployed applications, ensuring seamless integration with the DevOps workflow.

Configured, monitored, and optimized services for the Microsoft Azure platform to improve performance and scalability.

Ensured seamless integration of code changes across multiple environments.

Used Jenkins, Bamboo and Azure Pipelines to create CI/CD pipelines for .Net applications and APIs

Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl

Integrated AKS clusters with Azure Active Directory, Azure Key Vault, Azure Monitor, and Azure Policy for identity management, secrets management, monitoring, and governance

Optimized the performance and reliability of AKS clusters by applying best practices such as resource allocation, scaling, health checks, and logging

Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML

Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.

Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0

Conducted server and security audits, ensuring compliance with security policies and implementing necessary remediation measures. Used JIRA, Bugzilla and Remedy for bug tracking tools

Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases

Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.

Conducted regular infrastructure assessments to identify configuration issues, security vulnerabilities, and potential risks.

Developed PowerShell scripts and automation tools for managing and maintaining Azure infrastructure.

- Maintained and optimized Active Directory, Azure, MS Defender Suite, and M365 Environments for efficient user management and data security.

Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)

K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

- Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia

Healthcare Industry: Azure DevOps Engineer/Data Engineer/Python/R Developer/AI/ML/SP Consultant /DBA

Responsibilities:

- Managed automation technologies and enhanced build, test, and deployment capabilities using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Implemented IaC solutions using Bedrock and integrated them into CI/CD pipelines for efficient deployment.
- Contributed to the development of applications in Java, Microsoft .NET, and Oracle-deployed applications, leveraging DevOps best practices.
- Optimized services on the Microsoft Azure platform to ensure high availability and reliability.
- Built and deployed Docker containers for various web applications using Azure Container Registry and Azure Container Instances
- Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms

- Utilized Apache Tomcat, NGINX and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Troubleshot and resolved issues related to hardware, software, network, security, and performance
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Monitored and troubleshooted Azure infrastructure and application issues, ensuring high availability
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v3.0
- Automating systems administration by various automation scripting like Python, PowerShell, Bash, Puppet, etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Azure DevOps Engineer/Data Engineer/SharePoint Consultant /DBA
Responsibilities:

- Spearheaded the development of automated build and deployment pipelines using Azure DevOps, YAML, ARM Templates, and PowerShell to streamline development workflows.
- Designed and implemented Infrastructure as Code (IaC) solutions using Terraform, enabling consistent and reproducible deployments.
- Collaborated with cross-functional teams to develop and maintain DevOps CI/CD pipelines for seamless application delivery.
- Optimized cloud services on the Microsoft Azure platform to enhance performance and scalability.
- Designed and implemented network infrastructure using Cisco routers, switches, firewalls, and VPNs
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Conducted server and security audits, ensuring compliance with established security policies and implementing necessary controls. Documented network architecture, configuration, policies, procedures, and best practices
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR) option, etc.
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Azure DevOps Engineer/Data Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Offered technical assistance and mentorship to engineers, resolving issues promptly and efficiently.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Effectively maintained and improved AKS clusters in a production environment, ensuring seamless operations.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Automate build, test, and deployment processes using Azure DevOps, YAML, ARM Templates, and PowerShell.
- Implemented Infrastructure as Code (IaC) solutions using Terraform and Bedrock to automate infrastructure provisioning.
- Worked closely with development teams to integrate CI/CD pipelines into the software development lifecycle.
- Managed and optimized cloud services on the Microsoft Azure platform to ensure efficient resource utilization and cost-effectiveness.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR1, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.